

Mendel University in Brno
Faculty of Forestry and Wood Technology
Department of Engineering

**Development and Establishment of European Forest Machine Operator
Qualification Standards**

Diploma Thesis

2021/2022

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DIPLOMA THESIS TOPIC

Author of thesis: William Robb, BSc (Hons)
Study programme: European Forestry
Topic: Development and Establishment of European Forest Machine Operator Qualification Standards
Length of thesis: 50 pages

Guides to writing a thesis:

1. Get to know the issue of European Forest Machine Operator Qualification Standards. Compare newly developed standards and standards already used in other countries. Present your findings in your diploma thesis with regard to the frequently discussed issues covering this topic.
2. Develop European Forest Machine Operator (EFMO) learning outcomes based on performance criteria & underpinning knowledge in the operation of tractors, forwarders and harvesters in forestry.
3. Align the new EFMO learning outcomes with the European Qualification Framework.
4. Establish EFMO European occupational standards that can be used to develop qualifications for use by an appropriate international awarding body.
5. Perform a pilot test of EFMO European occupational standards in practice and evaluate the feedback you receive.
6. Work out your diploma thesis in compliance with the Directive LDF MENDELU no. 13/2019 "On the Form of Written Theses and Citations from Documents Used in Qualification Theses Presented at the Faculty of Forestry and Wood Technology". Consult the accomplished sections regularly; follow the terms predetermined by your diploma thesis supervisor.

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Diploma thesis topic submission date: November 2021

Deadline for submission of Diploma thesis: March 2022

L. S.

Electronic approval: 10. 1. 2022

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Acknowledgements

Special thanks to Ing. Tomáš Zemánek, Ph.D. for his dedicated support, understanding and consultation provided, prior to and during, the writing of this thesis.

Thanks also to the Erasmus + EFMO project for support and financial assistance ensuring thorough field testing was undertaken.

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1. Introduction

Numerous studies have identified a growing need for skilled forest machine operators throughout Europe (particularly harvester and forwarder operators). Recent European Union (EU) policy shifts in favour of the sustainable use of wood-based resources for bioenergy such as, an increase in the production and use of biomass crops, will ensure the demand for skilled forest machine operators remains high well into the future.

Associated with this increasing demand for skilled forest machine operators is a reported EU-wide increase in harvesting machinery accidents and fatalities. Fatalities include crush injuries due to machine overturns or unexpected movement etc, whereas injuries may result from repetitive movements, falls from the machine or fatigue factors. Over time, there has been an increasing tendency for forest owners/managers to engage the services of forest machine contractors. The increasing number of forest machine contractors (mainly private entrepreneurs') has been identified as a potential risk group.

Forestry in general is an industry sector where operator transnational mobility is very high across Europe. At the beginning of this study, forest machine operators did not have standardised vocational qualifications available at a European level. A variable range of training courses leading to national qualifications was available in some countries. Indeed, in some countries across Europe it is possible to gain employment as a forest machine operator without any specific proof of training or assessment of competency. Due to the high mobility of machine operators even into a neighbouring country in times of forest catastrophe such as wind or ice storms or the current bark beetle outbreaks, there is a need for an easily recognised Pan-European qualification, which is underpinned by common minimum occupational safety standards that a forest owner/manager or an employer can easily recognise and understand, thereby ensuring health and safety compliance has been assessed and met.

It is well known that a high company safety culture can be enforced through the impact of quality education and vocational training amongst other

factors. A key here is the notion of risk perception. In this context, forest machine operators have been known to expose themselves to inappropriate risks therefore this forms an important part of quality awareness raising training. Particularly as expensive, modern & highly specialised forest machinery and associated equipment and their operators are working within a potentially hazardous working environment. Only through increased access to training opportunities assessed and evidenced by certification can transparency of skills and qualifications in this sector gain recognition across Europe.

Further, improved recognition of forest machine operators' qualifications can support the overall international competitiveness of the European labour force on a global level and effectively reduce any potential barriers to employment & transnational movement of labour.

2. Aims and Objectives

The overall aims of this thesis are to investigate the current situation regarding forest machine operator qualification standards (where they exist), develop and establish European Forest Machine Operator (EFMO) occupational standards that can be used as the basis for European qualifications.

The individual objectives include as follows:

1. Research the issue of European Forest Machine Operator Qualification Standards and present the findings. Undertake a literature review to ascertain the current position in this field.
2. Develop European Forest Machine Operator (EFMO) learning outcomes based on performance criteria & underpinning knowledge in the operation of tractors (skidders), forwarders and harvesters in forestry. This includes an investigation of the required job tasks and level of knowledge to work safely. The learning outcomes will be mapped against any previous relevant findings.
3. Align the new EFMO learning outcomes with the European Qualification Framework (EQF). Investigate the EQF processes and how they can be applied to provide an indication of the appropriate levels of the new EFMO learning outcomes.
4. Establish EFMO European occupational standards that can be used to develop qualifications for use by an appropriate international awarding body. This will include completion of reviews and finalising the draft standards ready for use through engagement with training providers within Europe.
5. Perform a pilot test of EFMO European occupational standards in practice and evaluate the feedback received. Verify with training providers the final version of the standards for use in training and assessment of EFMO candidates. Finalise, live test and place on the market.
6. Complete the diploma thesis in compliance with the Directive LDF MENDELU no. 13/2019 "On the Form of Written Theses and Citations from Documents Used in Qualification Theses Presented at the Faculty of Forestry and Wood Technology" and follow the terms predetermined by the diploma thesis supervisor.

3. Literature review

Until now, Pan-European qualifications available in the field of forest machine operations have not existed. A key goal of this thesis has been to provide a concrete solution to the aforementioned issue in support of the safety and mobility of forest machine operators.

At the EU level, forestry and logging accounted for almost 0.2 % of the total GDP in 2019 and employed about 517 000 people (European Commission 2021a). European Commission (2021a) states that the area of forest in the EU increased by almost 10 % in 1990–2020 with the largest relative increase in Ireland (by 69 %). This can be compared with a very modest increase in countries such as Czech Republic (Fig. 1). Due to a perceived growing demand for skilled forest machine operators (Fobia 2019), Ireland was selected in this study, as a potential testing ground for a new Pan-European qualification in this sector. Additionally, forest mechanisation in Ireland as well as the UK and Nordic countries is near to 100% (Antti 2011).

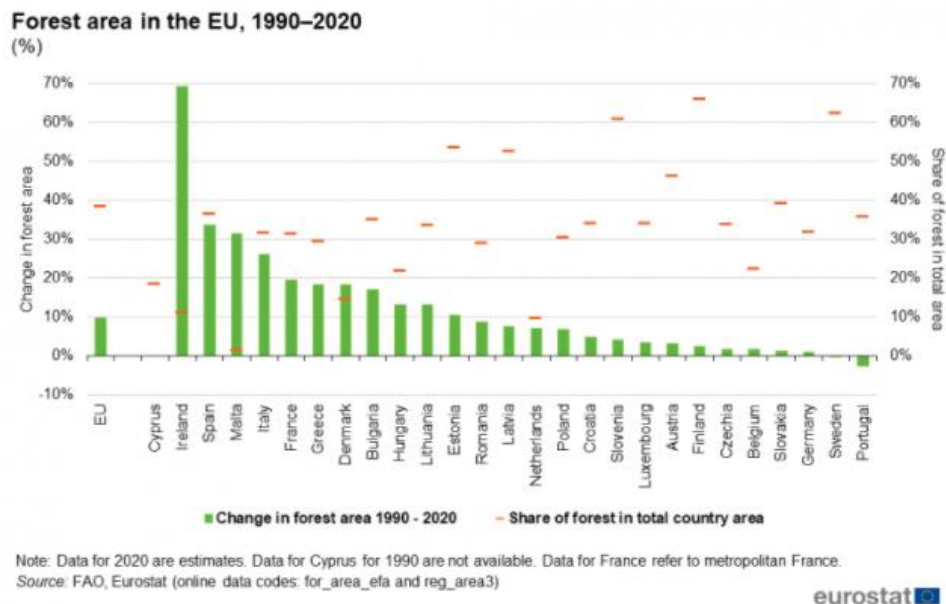


Figure. 1 – Comparison of EU forest areas (European Commission 2021a)

As highlighted in the introduction, recent European Union (EU) policy movements favour an increase for example, in the production and use of biomass crops ensuring a demand for skilled forest machine operators. A

major driving force currently at policy level within the EU, is the European Green Deal (2019) which states in relation to education and training, proactive re-skilling and upskilling are necessary to reap the benefits of the ecological transition. Indeed, preserving and restoring ecosystems and biodiversity are part of the key elements of the European Green Deal (Fig. 2).

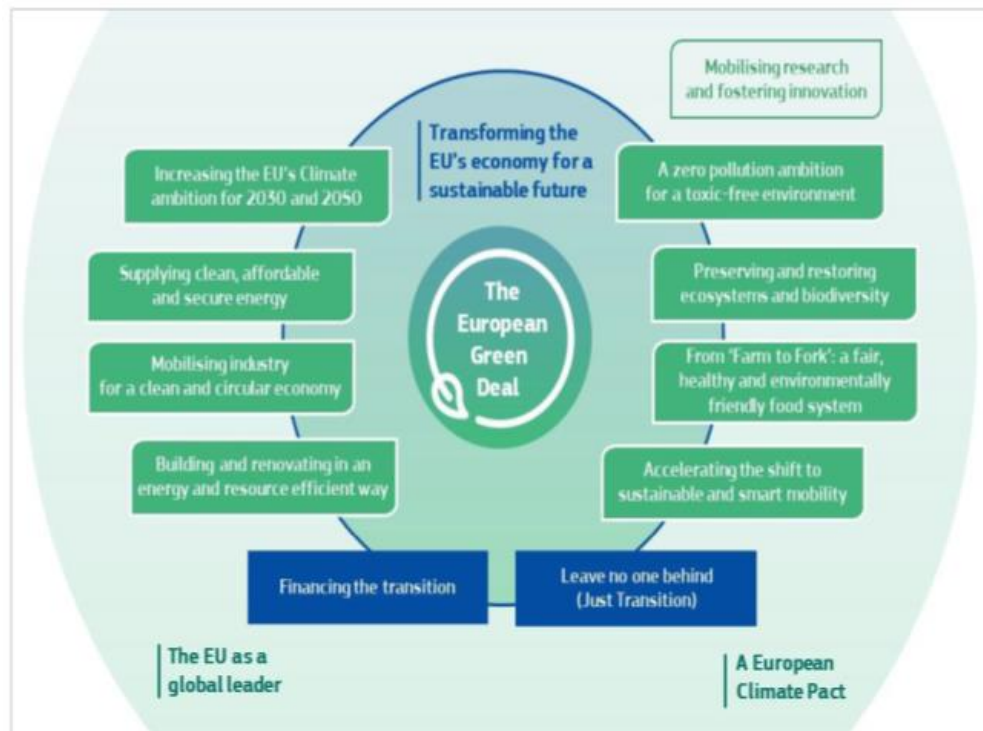


Figure. 2 – European Green Deal (European Commission 2019)

Anchored in the European Green Deal is the related EU 2030 Biodiversity Strategy 2030 (European Commission 2021b) and the New EU Forest Strategy for 2030 (European Commission 2021c). Both strategies in relation to forestry have a certain amount of overlap between them. For example, the European Commission (2021c) states that the forest-based sector holds significant economic potential for improving its production of sustainable and legally harvested wood which includes a range of forest machine operations.

The European Commission (2021b) states that a biodiversity-friendly increase in afforestation and reforestation will result in an increase in forest biomass for energy production. This will lead to an increased demand in skilled forest machine operators across Europe. This will include typical forestry harvester and forwarder operations (Figs 3 & 4).



Figure. 3 – Observation of forestry harvester operations in Czech Republic (ABA 2019)



Figure. 4 – Observation of forestry forwarder operations in Ireland (ABA 2019)

Training in forest machine operations where it exists throughout Europe is known to be highly variable and when forest owners or managers are

presented with a foreign certificates commonly, they are unable to ascertain the individual's competency level. It is known that in some European countries a formal certification does not exist and where they do exist, the standards required can vary, as confirmed during interviews (A Cach & M Lazar 2022, personal communication, 12 March). In effect, this leaves forest owners or managers open to the risk that the knowledge and competency of a forest machine operator is unknown. This can result in barriers to employment within a highly mobile European labour force due to a lack of understanding and recognition of a forest machine operators existing certification or where the certification is perceived not to meet the minimum safety standards of a particular country, as confirmed during interview (S Peoples 2020, personal communication, 8 October). This represented a need to develop formal qualifications in this sector based upon a set of minimum assessment standards that can easily be benchmarked across Europe. Even more so, because as stated by (Klun and Medved, 2007) in spite of technological advances, forestry work is still one of the most dangerous occupations especially when the training for operators is not adequate.

Provision of information and adequate training for workers are requirements of European legislation. Council Directive 89/391/EEC – measures to improve the safety and health of workers at work (EEC 1989) is the framework directive. It is supported by individual directives that address specific health and safety issues, of which, two are particularly relevant to forest machine operations. The Machinery Directive 2006/42/EC sets out the essential health and safety requirements that should be satisfied in order to ensure that machinery is safe which includes information and instruction (EU 2006). In accordance with the Work Equipment Directive 2009/104/EC, employers must ensure that workers given the task of using work equipment, receive adequate training (EU 2009). In the UK an independent assessment (separated from training) of a forest machine operators' competency is the recommended way of proving this (FISA 2013). This can be validated by a suitable national certification or other certification deemed to be equivalent. For example, an international certification undertaken to the same standards as the UK or higher. An international body which provides certification in this sector is ABA International (ABA) which has been assessed by the UK Health and Safety

Executive (HSE). The ABA European chainsaw certification was deemed by HSE to have similar arrangements in place to other scheme providers operating to national UK standards, as confirmed by email (A Hodgkinson 2021, personal communication, 28 September). The UK HSE is the industry regulator and National Occupational Standards (NOS) are managed by the government agency Skills Development Scotland (2021a), they are reviewed usually every 5 years by standard setting organisations and set at an equivalent level to the European Qualifications Framework (EQF) level 4. Forest machine operation standards evaluated include forest harvester operations (Tab. 1).

Table. 1 – UK forest machine standards-harvester (NOS LANTw38 2017)

Outcome	Occupational Standards-Forestry Harvester (UK)
1	Assess the risks associated with the site and the proposed work
2	Select and implement appropriate working method in accordance with the assessed risks
3	Select and use appropriate personal protective equipment (PPE) for the work
4	Maintain the security of machinery and equipment on site
5	Select, prepare and use the appropriate tools and equipment, safely and effectively
6	Confirm all equipment has been checked and is fit for purpose
7	Carry out routine operator maintenance, pre-start checks and the setting of the machine for use in felling trees
8	Select, prepare and maintain access and egress routes in a serviceable condition
9	Ensure that damage to the worksite, any remaining standing trees, tracks, roads, drains and the wider environment is kept within specified limits
10	Identify the trees to fell and position the machine to fell trees in the required direction, for both clear-fell and thinning operations
11	Select a felling method that is relevant to the tree size and condition and to the given specification
12	Fell trees mechanically, safely and effectively according to the specification

13	Position timber appropriately for subsequent processing
14	Meet given specifications for stump height (and stump treatment, where applicable)
15	Ensure the site is left in a condition which meets environmental requirements, in accordance with the specification
16	Maintain the health and safety of yourself and others at all times, in accordance with relevant legislation
17	How to identify hazards and assess risks
18	How to interpret risk assessments
19	The selection, use and care of personal protective equipment (PPE)
20	Site procedures including communication systems and emergency planning
21	The types of tools and equipment required and how to maintain and use these safely and effectively in line with the manufacturer's recommendations
22	The legal requirements for checking equipment
23	Why it is important to maintain the safety and security of equipment and vehicles on site
24	The current guidelines on machinery operation, risk zones and safety clearances from overhead electricity conductors, working at height and what to do in the event of contact with power lines
25	The function of all operating controls for base unit, machine boom and felling head, including the fire control system
26	The capabilities and limitations of the machinery used including its slope limitations
27	how to recognise the trees to be felled mechanically and how to select the size and species to meet the required specification
28	The implications of felling double, shattered or other malformed trees
29	How to deal with windblown trees with over-size, buried or inaccessible butts
30	The legal constraints and requirements for felling trees mechanically

31	How to check, maintain and sharpen the felling mechanism and the procedures to ensure the operator is safeguarded
32	The potential impact of your work on the environment and how this can be minimised
33	Your responsibilities under relevant environmental, health and safety legislation and codes of practice

The UK, Czech and Slovenian systems were researched in the case of the European Forest Machine Operator (EFMO), as it was hypothesised, to encourage quality standards of training to ensure candidates are able to pass an independent assessment. Additionally, prior to obtaining the harvester standard a candidate would need to have obtained a base machine standard which would require the successful achievement of 54 learning outcomes therefore together totalling a requirement to pass 87 learning outcomes in harvester operations. In comparison, although occupational standards for forest machine operators exist in Czech Republic these are more generic and not compulsory (Tab. 2). Specifically for harvester operations there are a total of 17 learning outcomes but these are grouped with forwarder operations which includes an additional 3 learning outcomes. Forwarder operations in the UK NOS (NOS LANTw47) contains 36 learning outcomes in comparison.

Table. 2 – Czech forest machine standards-harvester (NOS NSK 41-091-M 2020)

Outcome	Occupational Standards-Forestry Harvester (CZ)
1	Describe the principles of occupational safety and health in the movement of machines in the workplace and in the performance of their activities with regard to terrain and climatic conditions
2	Describe the main principles of workplace organization and occupational health and safety (workplace designation, hazardous area, landfills)
3	Describe the tasks necessary for the transport of machines to the workplace, their securing and the principles of transport of these machines

4	Describe and demonstrate the operational inspection of the machines before starting the activity
5	Identify basic forest tree species
6	Describe the function of a multi-operation harvesting machine, describe the controls and individual parts of its hydraulic arm and felling head
7	Describe the control of the hydraulic arm of the harvesting machine
8	Demonstrate the function of the hydraulic arm and felling head of the harvesting machine
9	Characterize the basic assortments of wood produced by the harvesting machine and demonstrate their production and classification
10	Enter the wood assortment data in the computer program of the logging machine control unit
11	Demonstrate felling and processing of felled trees
12	Describe the installation of tracks and wheel chains
13	Replace cutting bar, chain and hydraulic hose
14	Describe the maintenance tasks of harvester machines and perform basic maintenance of the machine according to the operating instructions
15	Demonstrate work with the operating instructions and service manual
16	Calibrate the measuring device
17	Fill in records of machine operation, including records of repairs and inspections performed

The NOS are managed by the Ministry of Agriculture, reviewed usually every 5 years and set at EQF level 4 (NOS NSK 41-091-M 2020).

Compulsory requirements to work with forest machines in Czech Republic include the possession of a driving licence for an agricultural tractor (the same requirement in the UK for public road use) and a certificate for working with a hydraulic crane (included in the base machine standards required in the UK). Training provided by the harvester or forwarder manufacturer is recommended (which is often omitted), as confirmed during

interview (T Zemanek 2022, personal communication, 14 March). A similar situation exists in other European countries as confirmed during interview (A Cach & M Lazar 2022, personal communication, 12 March) and standards such as COR 833106 – Operator in logging and timber handling referred to in a study reported to ABA (Campu 2021). For example, in Romania there is no occupational standard for harvester or forwarder operator. For the forwarder, the machine operator is trained in accordance with the standards for forestry tractors/skidder and in accordance with the crane operator standard. The forestry tractor/skidder qualification is mandatory to be able to work in the forest. Furthermore, the machine must be ISCIR certified (an approval from the authority responsible for checking the lifting installations is also required). The examination is held in front of a 3-person examination board (one chairperson and two members). One of the 3 people may be the one who trained the candidate as confirmed during an interview (R Campu 2022, personal communication, 20 March).

Despite some inconsistencies across a number of European countries (absence of standards/qualifications, lack of uniformity in provision or differences in qualification prerequisite requirements etc) the key NOS cited for forest machine operations were mapped and used as a benchmark in the development of harmonised EFMO qualifications for wider use across Europe. As Skills Development Scotland (2022) state NOS are used to influence the content and development of qualifications and training courses either directly or by setting out the skills and knowledge that are required to be met.

Other relevant NOS reviewed included the following:

- NOS LANTw79: Prepare and operate a base machine with attachments (2017)
- NOS LANTw43: Extract wood and wood products using motorised machines (2017)
- NOS LANTw42: Extract timber using mechanically powered winches (2017)
- NOS LANTw46: Attach and detach timber onto a skidder (2017)
- NOS LANTw47: Operate grapple loader (2017)

Assessment of the NOS by awarding bodies is done on a pass/fail basis which fails to recognise the merit and experience of individuals i.e., all certification holders are deemed equivalent. This is an over simplification which fails to adequately inform the employer of an employee's actual competence level. Additionally, the qualifications evaluated fail to take into consideration the possibility of 'skills fade' and requirements to refresh or update over time. Skills fade became evident during the Covid-19 pandemic which can lead to safety issues in high-risk sectors such as forestry. Beck (2020) states that reduced training and a reduction in task-based activity (those who have extended time in furlough) will result in skills fade. As previously stated, in some European countries it is common to have a three-person examination board which are responsible for organising and setting the exam and presumably reducing the chances of personal bias creeping into exam decisions. They usually are representative of industry and education/training. In the UK and Ireland this role can be undertaken by one suitably qualified individual. These factors were considered in the development of the EFMO qualifications.

Nieuwenhuis and Lyons (2002) state that in response to a survey of forest machine contractors, over 45% of respondents reported that improvements to health and safety in the sector could be improved through quality training. As highlighted earlier, this is more likely when the candidate is required to undertake an independent assessment of competence, as seen in the UK. Forest machine contractors is the same occupation identified in another study, as a potential risk group (Axelsson 1998) therefore training programs should prioritise this group.

A study in Poland highlighted similar concerns to Czech Republic around the lack of qualified forest machine operators and a need for the development of qualifications and competences which in turn can make the forestry profession more attractive particularly to younger people (Rutkowska & Adamowicz 2018). In the same study, a survey of students identified that interest in vocational training was very high in forest machine operations (Fig. 5).

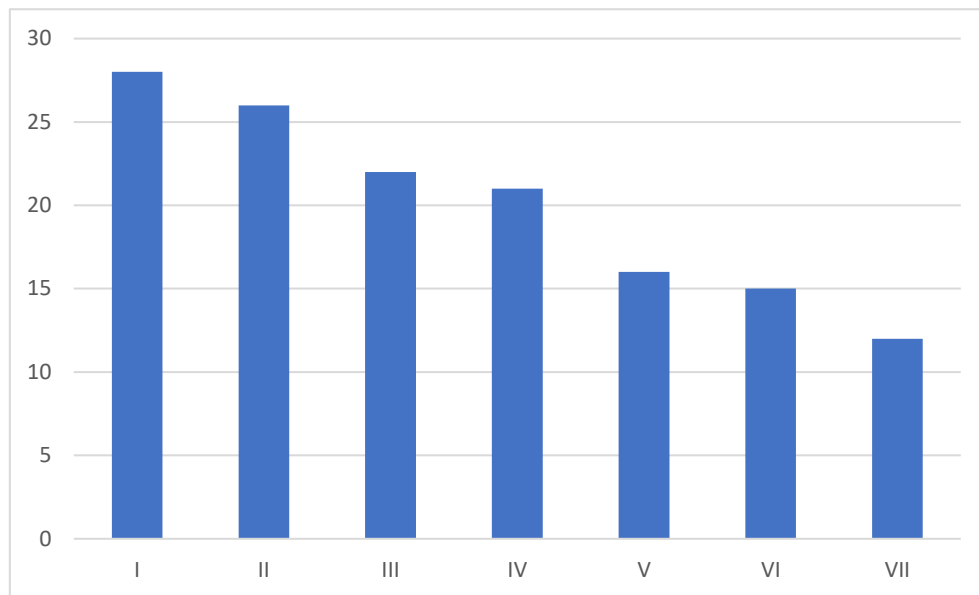


Figure. 5 – Interest in additional vocational training (Rutkowska & Adamowicz 2018)

I – forwarder operators – training and qualifications; II – harvester operators – training and qualifications; III – tree quality assessment – training and qualifications; IV – arborists – training and qualifications; V – wood sorting – training and qualifications; VI – forest thinning – training and qualifications, VII – stand tending – training and qualifications

The development of EFMO occupational standards providing training/assessment criteria would need to take into consideration and prioritise, the factors connected with fatalities and accidents, during forest machine activities. For example, during machine repairs (hydraulic failure); machine stability whilst working on different types of terrain (overturns/unexpected movement); machine entry and exit (slips/trips); machine transportation (boom contact with powerlines) etc. Additionally, Axelsson (1998) states that serious risks can be due to unintentional activation of photoelectric cells and switches. Factors such as musculoskeletal disorders, psychosomatic complaints and hearing loss are the most common health complaints which can lead to long-term work-related diseases (OSHwiki 2016). In harvesting operations with forest machinery in Germany, back pain is common amongst operators, as confirmed during an interview (M Lazar 2022, personal communication, 12 March). Chronic injuries such as repetitive strain injury (RSI) are relatively new and are related to the use of fully mechanised systems (Nieuwenhuis and Lyons 2002). Campu (2021) states the dangers associated with

tensioned winch cables, which frequently cause accidents, during the use of specialised forestry tractors known as skidders (Fig. 6).



Figure. 6 – A forestry skidder on a logging site in Romania (Campu 2021)

According to Wikipedia (2010) skidders are tracked or four-wheel drive tractors. Two common variants are a grapple version which lifts timber with a hydraulic bucket (claw) or a winch skidder which ‘skids’ the timber by a cable over the soil. They are more suited to thinning forests than forwarders and winch skidders particularly, are preferred for working on steep slopes in mountainous areas such as those found in Slovenia or Romania. Aalmo (2014) states that a gradient of 20–40% is where the transition between ground-based harvesting systems (wheeled or tracked machines) and cable-based timber extraction takes place. Steep and more difficult working conditions are likely to be a major contributory factor to accidents here. As stated in the introduction, machine overturns can lead to fatalities, this type of accident is likely to be more common on steep terrain due to stability issues. Another study identified that the use of tractors in Ireland accounted for over 58% of agricultural injuries between 2005-2014 (M Lee et al 2017). The agriculture sector includes forestry & logging. Here although the terrain is not generally very steep, in comparison to countries such as Romania, tractor overturns commonly occur due to other factors

which include rough terrain and loss of traction on banking's, drains, ditches etc (Fig. 7).



Figure. 7 – Overturned tractor on rough terrain (Dublin Fire Brigade 2021)

The EU Occupational Safety and Health (OSH) Strategic Framework on Health and Safety at Work 2021-2027, outlines 3 key priorities and actions for improving workers' health and safety (Fig. 8).

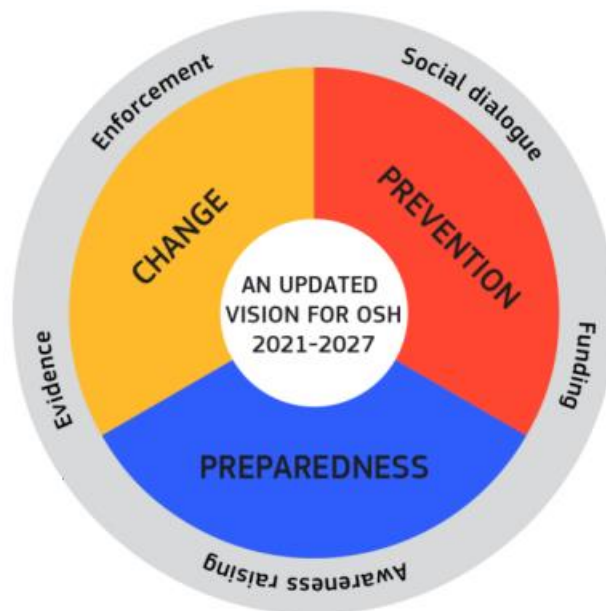


Figure. 8 – Vision for OSH 2021-2027 (European Commission 2021d)

EFMO occupational standards will support the EU-OSH key objective of improving prevention of workplace accidents and illnesses. This can be implemented through increasing forest machine operators' awareness of

risks during training. Additionally, the European Commission (2021d) states that OSH threats do not stop at borders and increased engagement with partner countries and international organisations can raise OSH standards worldwide. For this reason, the ABA International, a not-for-profit education and training association, was identified as a key partner and certification body in the delivery of the EFMO occupational standards.

Higher productivity can be achieved through developing and implementing work standards, and the skills of the workers can be improved through specific training (Aalmo and Talbot 2014). A survey undertaken by European Centre for the Development of Vocational Training (Cedefop 2015) highlights that there are significant economic and social benefits of education and training yet, 53% of the skilled forestry occupation group have low or no qualifications. It is clear that the development of Pan-European qualifications for EFMO can provide a number of benefits particularly if they are developed on foundation occupational standards which can be referenced to a European framework.

In relation to occupational standards the International Labour Office (ILO) provides an international standard classification of occupations. Here all occupations in the world are classified into groups according to similarity of skills levels and specialisation. Unit group 8341 represents mobile farm and forestry plant operators (ILO 2012). Forestry harvester, forwarder and tractor operations are occupations classified within this group. The system of classification applied at the occupational level is quite generic and does not reflect individual experience or qualification levels. This means for example, there is no distinction between a highly experienced harvester operator capable of undertaking complex tasks and a novice operator who lacks the ability to undertake complex tasks. Therefore, the development of EFMO qualifications should not completely reflect this generic classification but rather, consider a more specific and accurate model of distinction. However, a useful reference point for the development of occupational standards begins with the relevant ILO task descriptions (Tab. 3) for unit group 8341 (unrelated farm activities have been omitted from the list).

Table. 3 - Unit group 8341 task descriptions (ILO 2012)

Mobile forestry plant operators job tasks	
1	Driving and tending tractor-drawn or self-propelled special purpose forestry machinery to clear land, plant, harvest and carry trees or timber or perform other forestry operations
2	Preparing and positioning plant for operation
3	Adjusting speed, height and depth of implements
4	Operating plant to hold, lift and cut trees
5	Operating attachments to lift, swing, release and sort trees and logs, and operating auxiliary plant such as chipping machines and log splitting machines
6	Feeding felled trees into processors to strip limbs and cut into logs and loading logs onto stockpiles and into trucks
7	Servicing machinery and performing minor repairs

ILO (2012) also defines occupations within four broad skills levels which relate to the complexity and range of tasks undertaken within an occupation. Factors used to measure and categorise these may include the type of work undertaken by someone, their level of education, training and experience required. Forest machine operators are classified at skills level 2 (Tab. 4). This broad skills classification would need to be more specifically defined as highlighted previously in the development of EFMO occupational standards which should be measured in accordance with the actual competencies required, in the assessment of the final qualification.

Table. 4 - Skills levels comparisons between occupations (ILO 2012)

Skills level	Occupation examples
1	Office cleaners, garden labourers, freight handlers
2	Bus drivers, secretaries, police officers, motor vehicle mechanics, forest machine operators
3	Medical laboratory technicians, computer support technicians, shop managers
4	Civil engineers, secondary school teachers, medical practitioners

The European Green Deal (2019) mentioned previously as part of its skills agenda supports a similar type of reference framework for classifying occupations. It is referred to as the European Skills, Competences, Qualifications and Occupations (ESCO) it contains useful tools for describing learning outcomes etc (ESCO 2022). It also provides a link to a qualification database for Europe (Europass 2022). Here a search can be undertaken to find courses in the EU. A search for forest machine courses in Czech Republic revealed that only full-time, part-time etc school programmes were listed but not the specific vocational course sought. A further search found the qualification based on the forest machine standards previously mentioned (NOS NSK 41-091-M), on the website of the National Pedagogical Institute of Czech Republic (NPI 2022). The detailed standard could be found on the Czech version of the website and subsequently translated to English language.

Another useful reference in the development of EFMO standards is the EQF. As previously highlighted EFMO standards even if indicatively referenced to a European framework can aid qualification transparency. For example, where required to map against existing standards in different countries. The EQF was established in 2008 as a common reference framework of qualifications, expressed as learning outcomes at increasing levels of proficiency (European Commission 2018). It is important to note that the EQF does not harmonise or provide automatic recognition of qualifications between member states (European Commission 2018) but nevertheless is a useful tool in standards development for awarding bodies of qualifications. Quality assurance is core to the credibility of any qualification to which EQF provides guidance and recommendations (Tab. 5). These 10 well-established principles can easily be applied to international as well as national awarding bodies.

Table. 5 – Quality assurance principles for qualifications with an EQF level (European Commission 2018)

10 Quality assurance principles that should	
1	address the design of qualifications as well as the application of the learning outcomes approach

2	ensure valid and reliable assessment according to agreed and transparent learning outcomes-based standards, and address the process of certification
3	consist of feedback mechanisms and procedures for continuous improvement
4	involve all relevant stakeholders at all stages of the process
5	be composed of consistent evaluation methods, self-assessment and external review
6	be an integral part of the internal management, including sub-contracted activities, of bodies issuing qualifications with an EQF level
7	be based on clear and measurable objectives, standards and guidelines
8	be supported by appropriate resources
9	include a regular review of existing external monitoring bodies or agencies carrying out quality assurance
10	include evaluation results that can be accessed electronically

As well as applying the above principles, for ease of comparison of a qualification, whether or not it has an EQF level or similar, the qualification needs to be constructed of what are known as learning outcomes. The EQF definition of a qualification is the formal outcome of an assessment and validation process gained when a competent body determines that an individual has achieved learning outcomes (knowledge and skills & competence) to given standards (European Commission 2018). Learning outcomes provide a basis for descriptors which can be measured on 8 different levels (Tab. 6).

Table. 6 – EQF qualification levels part A (European Commission 2018)

Level	Knowledge	Skills
The complexity of the learning outcomes increases along with the	In the context of the EQF, knowledge is described as theoretical and/or factual.	In the context of EQF, skills are described as cognitive (involving the use of logical, intuitive and creative thinking)

qualification levels.		and practical (involving manual dexterity and the use of methods, materials, tools and instruments).
1	Basic general knowledge.	Basic cognitive and practical skills required to use relevant information in order to carry out tasks and solve routine problems using simple rules and tools.
2	Basic factual knowledge of a field of work or study.	A range of cognitive and practical skills required to accomplish tasks and solve problems by selecting and applying basic methods, tools, materials and information.
3	Knowledge of facts, principles, processes and general concepts in a field of work or study.	A range of cognitive and practical skills required to accomplish tasks and solve problems by selecting and applying basic methods, tools, materials and information.
4	Factual and theoretical knowledge in broad contexts within a field of work or study.	A range of cognitive and practical skills required to generate solutions to specific problems in a field of work or study.
5	Comprehensive, specialised, factual and	A comprehensive range of cognitive and practical

These levels can also be used as a more specific breakdown of the ILO forest machine operators' skills level previously highlighted and can be divided into 2 parts. This can further inform and enable the development of EFMO occupational standards. The levels are also designed in a way that a credit rating can be applied, as the levels increase, they become more complex and the learning time takes longer. Additionally, they are designed for use not only for vocational qualifications such as EFMO but they also cover academic and professional qualifications.

Generally, it appears that the forest machine operator's knowledge and skills could easily fit with level 3 although, when thinking in terms of forest calamities and the issues encountered, this may well overlap into the level 4 category or higher on occasion. As outlined previously the forestry sector is one of the most dangerous industries to work in. This is an important aspect that does not appear to have been factored into either the ILO's skills levels or the EQF qualification levels. This overlap can also extend to situations where more senior operators have more responsibility on work sites which often can include supervisory roles of less experienced operators. Alongside knowledge and skills, levels of responsibility are also classified by EQF (Tab. 7). In this case, experienced highly skilled operators could be classified on level 4.

Table. 7 – EQF qualification levels part B (European Commission 2018)

Responsibility and autonomy	
Level	In the context of the EQF, responsibility and autonomy is described as the ability of the learner to apply knowledge and skills autonomously and with responsibility.
1	Work or study under direct supervision in a structured context.
2	Work or study under supervision with some autonomy.
3	Take responsibility for completion of tasks in work or study. Adapt own behaviour to circumstances in solving problems.

4	Exercise self-management within the guidelines of work or study contexts that are usually predictable, but are subject to change. Supervise the routine work of others, taking some responsibility for the evaluation and improvement of work or study activities.
5	Exercise management and supervision in contexts of work or study activities where there is unpredictable change. Review and develop performance of self and others.
6	Manage complex technical or professional activities or projects, taking responsibility for decision-making in unpredictable work or study contexts. Take responsibility for managing professional development of individuals and groups.
7	Manage and transform work or study contexts that are complex, unpredictable and require new strategic approaches. Take responsibility for contributing to professional knowledge and practice, and/or for reviewing the strategic performance of teams.
8	Demonstrate substantial authority, innovation, autonomy, scholarly and professional integrity and sustained commitment to the development of new ideas or processes at the forefront of work or study contexts, including research.

Whilst the EQF provides a useful reference point for course development it appears overall to be more suited to academic attainment rather than vocational. This viewpoint is substantiated to a degree by a technical critique of the EQF including identified issues and limitations (Lester 2015). Here the author points out that occupational qualifications (this includes EFMO) do not easily cross-reference to the levels. As Lester (2015) states this is due to essential content being left out or the complexity level placed incorrectly from what is undertaken in reality at the workplace. The workplace can be viewed particularly in terms of cognitive skills and vocational experience as more fluid than the academic environment.

Therefore, whilst useful tools in the development of EFMO both ILO and EQF they must not be followed too rigidly or the final qualifications may not completely reflect the actual needs of the forest industry from a practical vocational perspective. This point around qualification development and suitability in practice can be observed when undertaking mapping exercises of industry requirements and needs across to sub-levels of the EQF i.e. individual national levels where this information is readily available for the forestry sector.

From an international perspective, an organisation that provides useful supporting resources in skills development and training is the United Nations Educational, Scientific and Cultural Organisation (UNESCO). Under the umbrella of UNESCO, the body specifically responsible for this is the International Centre for Technical and Vocational Education and Training (UNEVOC). UNESCO (2021) states that at a certain point in the design of curricula and for assessment purposes, competencies need to be put into an observable and measurable form.

The same document goes further and supports the conclusions made by Lester (2015) as it ascertains by shifting the direction of training from state or macro-level to meso-level organizations (Fig. 9) training will reflect a better sense of the challenges that exist in the world of work (UNESCO 2021). This is part of what UNEVOC labels as the ‘three-I’ approach, whereby vocational education is the sector closest to the labour market where new qualifications can be quickly identified, integrated into the training environment and implemented. It should be kept in mind that this relates specifically to the sphere of technical and vocational education and training.



Figure. 9 – Components of the ‘three-i’ approach (UNESCO 2021)

As well as internal stakeholders, UNESCO (2020) states that during qualification development, an emphasis should be placed on the involvement of external stakeholders. For example, to awareness raising meetings, technical and multiplier events with representatives of interested parties. Relating to the development of EFMO this was the responsibility of the nominated body ABA International with access to funding support. Funding support was applied for in the form of an Erasmus + application in 2019 and was awarded. The need to engage with forest and forestry stakeholders, is also referred to within the new EU Forestry Strategy for 2030 (European Commission 2021b) which supports the parties working together to adapt education and training for foresters to the challenges and needs of today’s realities. An EFMO project led by myself (Project Coordinator) although to a certain extent impacted by the Covid pandemic, has managed to engage with a number of external stakeholders (Fig. 10) throughout Europe since 2019. Examples in Czech Republic (ABA 2022), have included a range of meetings, participation and endorsements from the following:

- PEFC Czech Republic which represents 250 Czech forestry organisations. Program for the Endorsement of Forest Certification Schemes (PEFC) is the world’s largest forest certification system.

- Merimex the sales and service representative of John Deere the worlds largest manufacturer of specialised forestry equipment
- Mendel University in Brno staff (Department of Engineering and Wood Technology)
- State forestry body Vojenské lesy a statky ČR (VLS)
- School and training facility of the Fire and Rescue Service (Hasici) of the Czech Republic (ŠVZ HZS ČR) of the Ministry of the Interior
- Czech Landscape and Garden Society (SZKT). A Civil Association with over 500 members
- NejLes sro a forest management, consultancy and harvesting business
- Secondary School Forestry Bzenec provider of forest machine operation courses
- All4trees sro a tree care business and training provider
- Skolení mp a forestry training provider



Figure. 10 – External stakeholders in qualification development (UNESCO 2020)

This literature review has set out the case and rationale, including the multiple benefits, that can be associated with harmonised Pan-European EFMO qualifications, especially improved safety standards and transnational mobility of workers.

The next step required a methodology in the development and establishment of EFMO occupational standards utilising tools such as the EQF and others, to enable the establishment of the ‘building blocks’ i.e., learning outcomes and associated criteria, which reflect forest machine operations and tasks in practice. Prior to this, as Pan-European occupational standards did not exist for this sector, a brief mapping exercise was undertaken reflecting the geography of the territory of Europe, to include Slovenia which has a more mountainous environment (NOS NPK-78281880 2017 and NOS NPK-6718282011 2018).

NOS mapping is important as Skills Development Scotland (2021b) states that work-based learning should be made efficient, responsive and future focussed. This principle was seen as integral to the development, long-term maintenance and management of Pan-European occupational standards.

4. Materials and Methods

4.1 Description

The main sources of information reviewed and analysed provided insight into the current state-of-the art, relating to forest harvesting machinery accidents and fatalities across Europe, standards of national training provision (specifically UK and Czech Republic), the need for EFMO occupational standards and availability of qualification development resources. In alignment with the aims and objectives, the process required the implementation of the following measures:

- Research the issue of European Forest Machine Operator Qualification Standards and present the findings (**Thesis aims & objectives no.1**)
- Undertake a cross-referencing exercise aimed at mapping typical European standards (**Thesis aims & objectives no.2**)
- Development of innovative EFMO learning outcomes based on performance criteria & underpinning knowledge, aligned with the EQF and established them as minimum occupational standards for use across Europe (**Thesis aims & objectives no.3**)
- Development EFMO qualifications and establish a certification scheme within the framework of the umbrella body ABA (**Thesis aims & objectives no.4**)
- Undertake training & standard setting for EFMO skills trainers & assessors (**Thesis aims & objectives no.4**)
- Undertake pilot testing of the new EFMO qualifications review and evaluate the results (**Thesis aims & objectives no.5**)
- Finalise and deliver live testing of the qualifications into the marketplace (**Thesis aims & objectives no.5**)

4.2 Thesis aims & objectives no.1

4.2.1 Research qualification standards and present the findings

This part of the thesis was mainly undertaken as a desk-top study including a literature review (secondary research) but also involved an element of field research (primary research). The key focus of the literature review was to research the current situation, via scientific publications, and other materials from previous research including undertaking an investigation of

suitable national forest machine qualification standards. Field research activities included engagement with forest machine operators, trainers, stakeholders etc by undertaking interviews and face to face meetings physically on-site whilst observing the activities or by other means of personal communication (e-mails, telephone, social media etc) and numerous focus group discussions (EFMO project). These additional activities helped to provide a better insight into the subject matter analysed and related issues. In total, over 50 documents were reviewed and 30 stakeholder meetings were held (not including numerous local meetings) with the support of the Erasmus + EFMO project.

4.3 Thesis aims and objectives no.2 and no.3

4.3.1 Standards mapping and data analysis

The priority here was to undertake a mapping exercise of national forest machine operator standards collated earlier. Following initial review and as highlighted previously, forest machine operation occupational standards at national levels in Europe, where available, were variable in content and delivery mechanisms. The aim was to ensure national information on vocational standards for forest machine operations that were obtained and collated would undergo data analysis. The qualitative methodology applied here consisted of three stages:

- Stage 1-was undertaken as a content analysis (reviewing and familiarisation) of the data particularly for relevance of subject matter
- Stage 2-required a comparative analysis in order to understand either the level of similarity or differences in the data
- Stage 3-involved a combination of thematic (patterns) and interpretive (meaning) analysis by observation and establishing the learning outcome patterns including the meaning of unique data characteristics

It was also important that the process of information collation was representative of the European situation from a geographical point of view. Where required basic translations were undertaken using the google translate tool and technical clarity sought as necessary from consultation with industry experts fluent in the required national language. The sample

countries were selected as they represented typical examples from north-western Europe through central Europe to south-eastern Europe. Namely the UK and Ireland, Czech Republic and Slovenia. This provided three data sets for analysis (Tab. 8). The sample size may seem small but the NOS at national levels has already been developed and coded therefore minimising any potential benefit from a larger sample size at this stage. It is possible that future studies with a larger sample size, mapped against the newly established Pan-European standards, may serve to further ensure that the competency levels remain fit for purpose.

Table. 8 – Data sets analysed (Robb 2019)

Country	United Kingdom
Data set 1	Harvester NOS
Standard Name and Code	Fell trees mechanically (code: LANTw38) & Prepare and operate a base machine with attachments (code: LANTw79)
Data set 2	Forwarder NOS
Standard Name and Code	Operate grapple loader (code: LANTw47) & Prepare and operate a base machine with attachments (code: LANTw79)
Data set 3	Skidder NOS
Standard Name and Code	Extract wood and wood products using motorised machines (code: LANTw43) Extract timber using mechanically powered winches (code: LANTw42) Attach and detach timber onto a skidder (code: LANTw46) Prepare and operate a base machine with attachments (code: LANTw79)
Country	Czech Republic
Data set 1 & 2	Harvester & Forwarder NOS
Standard Name and Code	Timber harvesting with logging and transport machines (code: 41-091-M)

Data set 3	Skidder NOS
Standard Name and Code	Logging operations part d (code: 41-023-H)
Country	Slovenia
Data set 1 & 2	Harvester & Forwarder NOS
Standard Name and Code	Operator / operator of machines for machine felling and export of wood with articulated semi-trailer (code: 78281880)
Data set 3	Skidder NOS
Standard Name and Code	Forestry tractor (code: 6718282011)

Once the mapping process was completed a qualitative analysis of the data was then undertaken, in accordance with the methodology outlined earlier. A learning outcome-based template was designed to facilitate this, in line with the EQF format, which related to performance criteria and underpinning knowledge. The main task here, of thesis objective 3, was following a methodical evaluation of each country's standards, the technical content could then be transposed into a best fit framework for developing the innovative Pan-European occupational standards. Performance criteria and underpinning knowledge were evaluated and cross-referenced where appropriate and allocated to a position in the new Pan-European framework, avoiding unnecessary duplication whilst potentially identifying any gaps.

4.4 Thesis aims and objectives no.4 and no.5

4.4.1 Development of new EFMO qualifications and testing

Once the new benchmark occupational standards were established, with the engagement of stakeholders, a basis for a certification framework was developed. Various stakeholder representatives were also invited to correspondingly establish an EFMO expert group. This would form a valuable advisory body containing subject matter experts. For example, as Ireland was selected as the testing ground for the new EFMO qualifications, it made sense to engage with the national stakeholder Coillte. According to Forest Europe (2020) forests of the State Forest agency, Coillte, which accounts for 51% of all forests, are certified by both FSC and PEFC, and account for most of the certified area. This would mean that the new EFMO qualifications would not only align (as a minimum) to any existing national

qualification standards but also fit with wider industry certification requirements.

Thesis objectives 4 and 5 set out that once the qualifications are established the standards had to be set and tested in practice which allowed any final adjustments to be made as necessary. Under the umbrella body ABA, the standards would be owned, maintained and set periodically, for international use. Standards setting can be considered as a form of quality assurance of the EFMO product to the consumers. The criterion-reference orientation links the standard to the content of the competence level under consideration (Ben-David 2000). The EFMO standards have been criterion referenced. Ben-David (2000) states that increased standard-setting enhances high-quality testing and raises confidence in the testing of professional competencies. EFMO standards were designed to be outcome-based. This meant that learning outcomes were clearly defined criteria that performance assessment decisions could be based upon. EFMO trainers and assessors contributed to the standard setting process and subsequent pilot testing. Initially, key stakeholders within the EFMO project were well positioned to hold local meetings, invite, select and involve suitable participants directly in the process (Fig. 11). This is complementary to the wider target group seen in figure 10 previously.

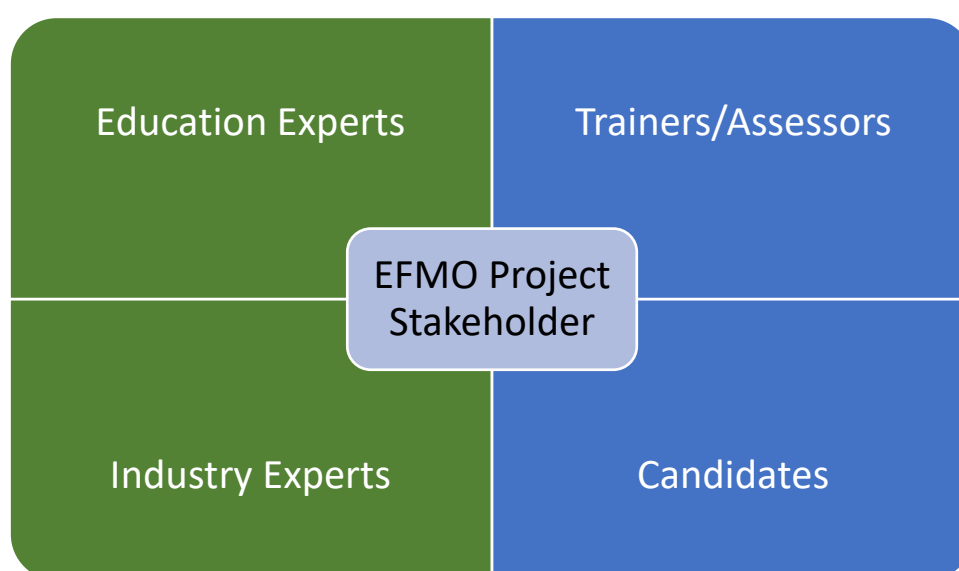


Figure. 11 – EFMO project stakeholder tasks a) establish expert group (green) and b) select trainers/assessors and candidates (blue) (Robb 2019)

The EFMO project stakeholder group included representation from across Europe as follows:

- A1 Arborists Ltd-United Kingdom (Project Applicant & Manager)
- Mendel University in Brno-Czech Republic
- Ogre Technical School-Latvia
- National Forest Centre-Slovakia
- SGLS Secondary Forestry School-Slovenia
- Coillte, State Forest Agency-Ireland
- Transilvania University of Brasov-Romania

A number of EFMO project management meetings and technical training events were held by the stakeholders prior to the onset of the pandemic in early 2020. The first of these was held by myself with the State Forest Agency in Ireland during October 2018 to promote the idea and discuss a way forward. This resulted in the initial EFMO standard setting and pilot tests taking place in Ireland early in 2019 with experienced registered ABA trainers/assessors who were independently assessed and verified (Fig. 12).



Figure. 12 – EFMO project trainers/assessors participating in standard setting and initial pilot tests in Ireland (Robb 2019)

The pilot tests were mainly undertaken at various locations throughout Ireland (Tab. 9). These involved not only candidates but all the stakeholders, trainers, assessors and the EFMO expert group. Following completion, these forest machine operators were later selected as ABA-EFMO qualification candidates and undertook live tests.

Table. 9 – Pilot tests-EFMO Forest machine qualifications (Robb 2019)

Pilot Test no.	Machine type	Test date	Country	Test duration (hours)	Participant no.
1	Harvester	21/1/19	Ireland	7	16
2	Forwarder	22/1/19	Ireland	6.5	9
3	Forwarder	24/1/19	UK	5.5	9
4	Harvester	8/4/19	Ireland	5.5	12
5	Forwarder	9/4/19	Ireland	5	9
6	Harvester	30/5/19	Ireland	3.5	7
7	Forwarder	31/5/19	Ireland	3.5	7
8	Harvester	25/6/19	Latvia	3.5	23
9	Forwarder	26/6/19	Latvia	5.5	19
10	Skidder	27/6/19	Latvia	4.5	19
11	Skidder	10/6/20	UK	4	14
12	Forwarder	5/10/21	Ireland	3	6
13	Forwarder	6/10/21	Ireland	3	6
14	Skidder	7/10/21	Ireland	4	10

Pilot tests were undertaken with both wheeled and tracked machines which included some of the most commonly used machines manufactured by John Deere, Ponsse, Komatsu etc. Once pilot test assessment durations became consistent (Fig. 13), standard setting was no longer required and the ABA trainer/assessors prepared and organised, for themselves to subsequently become the first candidates undertaking the live tests. Pilot test parameters in practice appeared to be a minimum of four with at least the last two assessment durations remaining consistent. Pilot testing was completed earliest for forestry harvesters and therefore became the first EFMO qualification to be live tested.

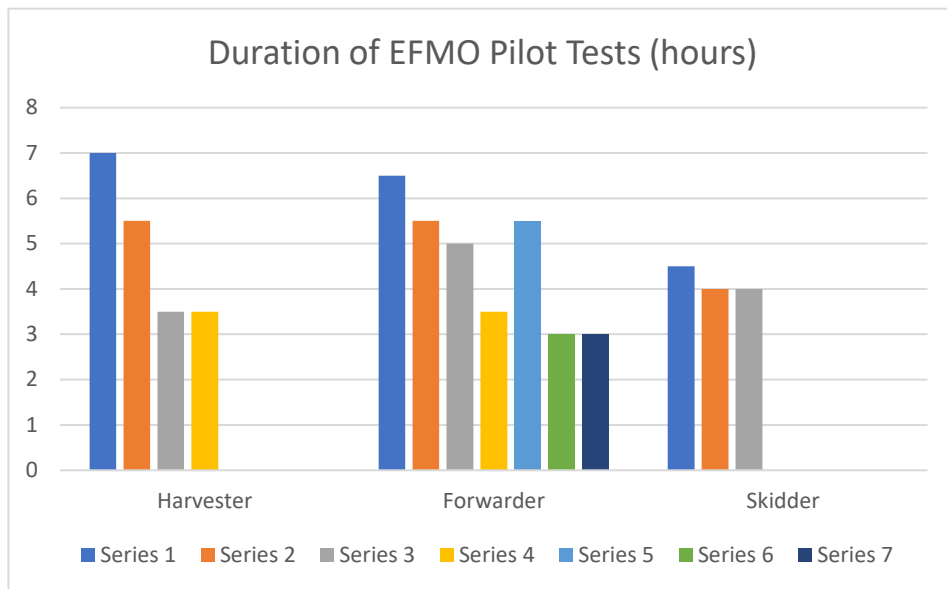


Figure. 13 –Number and duration of EFMO pilot tests in (Robb 2021)

65 live tests of candidates have since been undertaken and were primarily focused on harvester and forwarder operations, as skidding was not in demand, on the mainly flat terrain of the field testing sites in Ireland. Nevertheless, the base machine component was successfully field tested. Further field tests and revisions of the forestry tractor/skidder qualification aim to be completed by mid-2022. A minimum of at least one more pilot test is required here, depending upon the outcome.

5. Results

5.1 Thesis aims & objectives no.1

- Research the issue of European Forest Machine Operator Qualification Standards and present the findings (**no.1**)

5.1.1 Research qualification standards and present the findings

The literature review findings including primary and secondary research referred to in chapter 3, have been summarised in the following table (Tab. 10).

Table. 10 – Literature review findings (Robb 2022)

No.	Results
1	The demand for skilled forest machine operators throughout Europe is increasing
2	Forest harvesting machinery accidents and fatalities across Europe is increasing
3	Forest machine contractors (private sector/self-employed) represent the group of forest machine operators at highest risk of injury
4	On national levels where they exist, occupational standards are highly variable and European EFMO qualifications did not previously exist
5	There is a clear need to develop formal EFMO qualifications based upon a set of minimum assessment standards that can easily be benchmarked across Europe supporting worker mobility
6	EFMO occupational standards will support the EU-OSH key objective of improving prevention of workplace accidents and illnesses
7	EFMO qualifications can support and ensure employers meet their legal requirements
8	Increased access to EFMO training opportunities assessed and evidenced by certification can improve recognition of the sector in Europe
9	Improved recognition of EFMO qualifications can support the competitiveness of the European labour force and reduce any

	potential barriers to employment & transnational movement of labour
10	Numerous resources at both European and international levels can support and guide the development of EFMO occupational standards and qualifications such as the EQF
11	An EFMO (Erasmus +) project which includes engagement with stakeholders has been ongoing since 2019 to support the achievement of the aims and objectives of this thesis and long-term sustainability thereafter

5.2 Thesis aims and objectives no.2 and no.3

- Undertake a cross-referencing exercise aimed at mapping typical European standards (**no.2**)
- Development of innovative EFMO learning outcomes based on performance criteria & underpinning knowledge, aligned with the EQF and established them as minimum occupational standards for use across Europe (**no.3**)

5.2.1 Standards mapping aligned with the EQF and data analysis

For the reasons described previously the UK standards as seen in the harvester example (Tab. 1) were used as the benchmark to which the other selected standards were mapped against, which resulted in mapping document no.1 (Tab. 11). Table 8 outlined the three key data sets for subsequent analysis.

5.2.2 Data set 1: Forestry harvester analysis

The UK standard LANTW79-Prepare and operate a base machine with attachments has been integrated, for practical purposes, within the mapped standards to avoid unnecessary duplication of content (highlighted text and coded (b)). Currently the UK standards are also recognised in Ireland. Where an outcome box is left empty it means equivalent outcomes were not found in the corresponding NOS. The documents analysed and mapped here were as follows:

- NOS LANTw38: Fell trees mechanically (2017)
- NOS LANTw79: Prepare and operate a base machine with attachments (2017)

- NOS NSK 41-091-M: Timber harvesting with logging and transport machines (2020)
- NOS NPK-78281880: Operator / operator of machines for machine felling and export of wood with articulated semi-trailer (2017)

Table. 11 – Mapping document 1-Content analysis of UK, CZ and SI Forestry Harvester standards (Robb 2019)

UK Outcome	European Occupational Standards- Forestry Harvester (UK)	CZ Outcome	SI Outcome
PERFORMANCE CRITERIA-you must be able to:			
1	Assess the risks associated with the site and the proposed work	1	1
2	Select and implement appropriate working method in accordance with the assessed risks	2	1
3	Select and use appropriate personal protective equipment (PPE) for the work		2
4	Maintain the security of machinery and equipment on site		2
5	Select, prepare and use the appropriate tools and equipment, safely and effectively		2
6	Confirm all equipment has been checked and is fit for purpose (including attachments)		2
7	Carry out routine operator maintenance, pre-start checks (includes safety guards) and the setting of the machine for use in felling trees (includes operator environment)	4	2,3,7
8(b)	Check the immediate work area for hazards and obstacles before		3

	moving off (use appropriate warning signals at all times)		
9	Select, prepare and maintain access and egress routes in a serviceable condition		3
10(b)	Operate and manoeuvre the base machine safely, and in a way that is consistent with the type of vehicle, attachment and operation		3
11(b)	Make sure the attachment is in transport position when not in use, in accordance with specifications		3
12(b)	Modify operating procedures to take into account any changes in weather, ground conditions, types of terrain and site restraints		1,3
13(b)	Utilise additional safeguards when reversing or manoeuvring and comply with safety distances adjacent to roads and tracks or where others are working		3
14(b)	Maintain the efficiency of base machine and attachment performance through the appropriate handling and maintenance	1,4	3
15(b)	Deal with any hazards and obstacles encountered during the operation, in accordance with standard practice		3
16(b)	Monitor the operation of the attachment and recognise any faults		3
17	Ensure that damage to the worksite, any remaining standing trees, tracks, roads, drains and the wider		3

	environment is kept within specified limits		
18	Identify the trees to fell and position the machine to fell trees in the required direction, for both clear-fell and thinning operations		3 (less specific)
19	Select a felling method that is relevant to the tree size and condition and to the given specification		3
20	Fell trees mechanically, safely and effectively according to the specification	2	3
21	Position timber appropriately for subsequent processing		3
22	Meet given specifications for stump height (and stump treatment, where applicable)		3
23(b)	Stop and isolate the base machine on completion of the activity, in line with standard practice		
24(b)	Remove the attachment, where appropriate, in accordance with specifications		
25(b)	Leave the base machine and attachment secure after use and in a suitable condition for future use		
26	Ensure the site is left in a condition which meets environmental requirements, in accordance with the specification		9
27	Maintain the health and safety of yourself and others at all times, in accordance with relevant legislation		9
KNOWLEDGE & UNDERSTANDING-you need to know & understand:			

1	How to identify hazards and assess risks	1	9
2	How to interpret risk assessments		5
3	The selection, use and care of personal protective equipment (PPE)		9
4	Site procedures including communication systems and emergency planning		8
5	The types of tools and equipment required and how to maintain and use these safely and effectively in line with the manufacturer's recommendations	1	7
6	The legal requirements for checking equipment		3
7	Why it is important to maintain the safety and security of equipment and vehicles on site		2
8	The current guidelines on machinery operation, risk zones and safety clearances from overhead electricity conductors, working at height and what to do in the event of contact with power lines		1,3
9	The function of all operating controls for base unit, machine boom/ attachment and felling head, the fire control system, including on board computer system where fitted	2,3,4	2,3
10	The capabilities and limitations of the machinery used including its slope limitations		3
11(b)	The required pre-start checks and adjustments for the base machine	1	2

	and attachments, and the reasons why these must be carried out		
12(b)	The requirements for machine and operator protection structures		
13(b)	The procedure for loading and unloading base machine and attachments from transporter where appropriate	1	
14(b)	How to mount and dismount the base machine safely		
15(b)	How to start and stop the base machine including the correct procedures for jump starting		
16(b)	The types of attachments that are safe for use with the base machine and how they are fitted		
17(b)	The conditions that should be taken into account when considering the use of attachments		
18(b)	How to set or calibrate attachments, where required	4	
19(b)	The appropriate use of warning signals and indicators		
20(b)	The ways in which the base machine should be operated and manoeuvred, and how different weather, ground conditions, types of terrain and the work being carried out must be taken into account		1,3
21(b)	The various driving characteristics and techniques for driving a base machine with different wheel configurations		

22(b)	Correct fitting of traction aids and advantages and disadvantages of using them	2	
23(b)	How to operate and use attachments safely including the counterbalance		
24(b)	The controls and electrical services on the base machine and the attachment		
25(b)	How to put the attachment into transport position and work position, where appropriate	1	
26(b)	The capabilities of the base machine and the factors that may affect its efficiency		
27(b)	The types of hazards and obstacles that may be encountered and how these should be dealt with		1
28(b)	How to monitor the operation of the attachment and the actions that might be required		6
29(b)	The faults, defects and wearing of parts that occur with attachments used for treework operations		
30(b)	How to preserve the efficiency of the base machine and attachment through routine operator checks and basic maintenance in line with specifications	4	7
31(b)	How to carry out basic repairs and where to refer problems with the base machine or attachment	4	7
32(b)	The reasons why the base machine and attachment should be left in a suitable condition for future use		

33	How to recognise the trees to be felled mechanically and how to select the size and species to meet the required specification		1,3,9
34	The implications of felling double, shattered or other malformed trees		
35	How to deal with windblown trees with over-size, buried or inaccessible butts		
36	The legal constraints and requirements for felling trees mechanically		9
37	How to check, maintain and sharpen the felling mechanism and the procedures to ensure the operator is safeguarded		7 (less specific)
38(b)	How to set up and use fuelling and maintenance areas where appropriate		
39	The potential impact of your work on the environment and how this can be minimised		9
40	Your responsibilities under relevant environmental, health and safety legislation and codes of practice		9

The data obtained from the forestry harvester standards evaluation was compared to measure the level of similarity (Figs. 14 & 15).

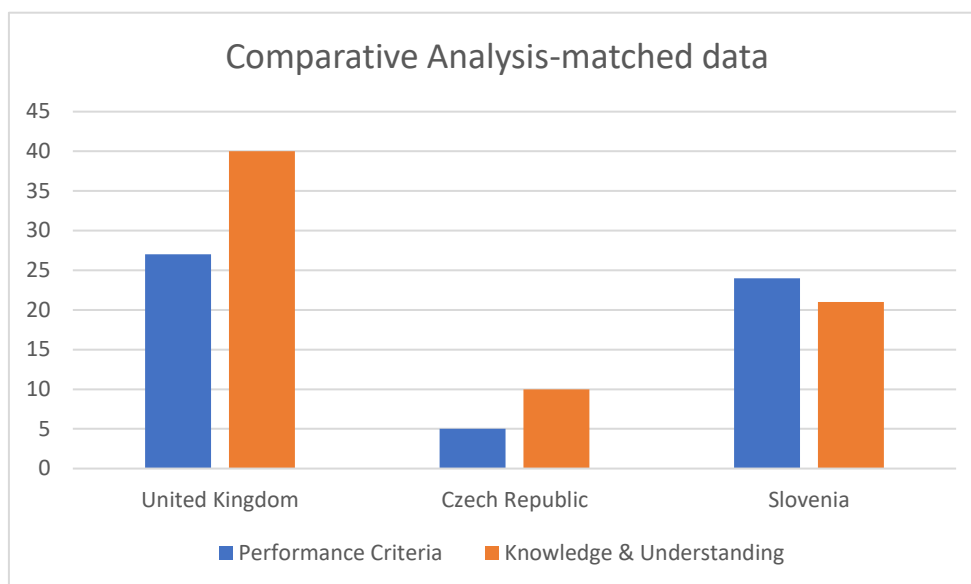


Figure. 14 – Evaluation of similarity between matched NOS learning outcomes in different forestry harvester standards (Robb 2019)

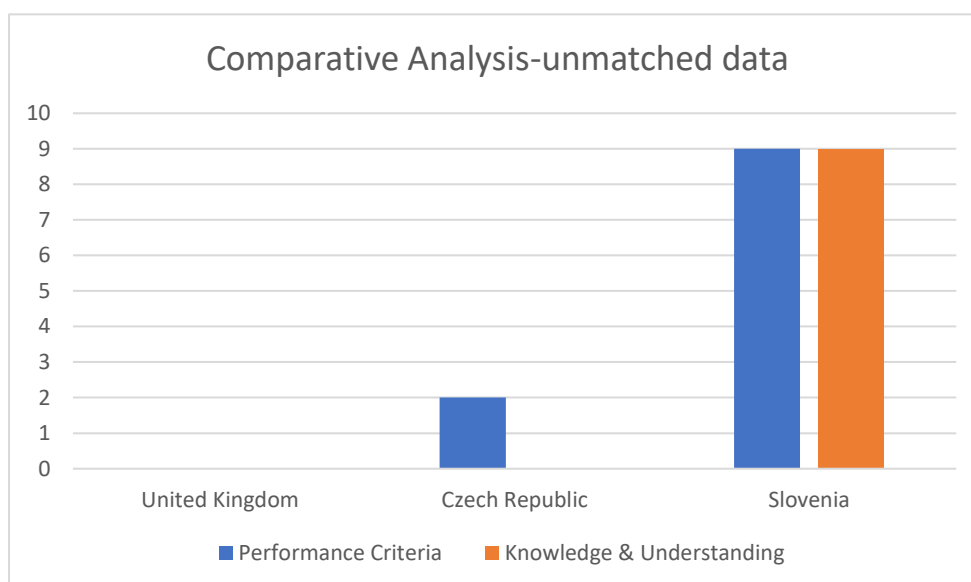


Figure. 15 – Evaluation of similarity between unmatched NOS learning outcomes in different forestry harvester standards (Robb 2019)

Additional learning outcomes beyond the scope of the benchmark UK standards were identified (Tab. 12).

Table. 12 – Mapping document 2-Comparative analysis of UK, CZ and SI Forestry Harvester standards-additional findings (Robb 2019)

CZ Outcome	European Occupational Standards- Forestry Harvester (CZ)	UK Outcome	SI Outcome
ADDITIONAL PERFORMANCE CRITERIA-you must be able to:			

2	Identify basic forest tree species		
4	Replace cutting bar, chain and hydraulic hose		
SI Outcome	European Occupational Standards- Forestry Harvester (SI)	UK Outcome	CZ Outcome
ADDITIONAL PERFORMANCE CRITERIA-you must be able to:			
1	Get acquainted with the tasks in the harvesting units and on the job site with the help of a work order, harvesting plan or contract		
1	Reads forestry maps		
2	Check the suitability and amount of fuel and lubricants in the machine		
3	Before felling, make sure there is no one in the danger area		
3	Determine the degree of damage inside the tree - the presence of rot		
4	Keep records: fuel/oil consumption, machine hours, faults, maintenance etc		4 (less specific)
6	Check the accuracy of the required assortment lengths and the correctness of assortment diameters		
8	Communicates with co-workers with gestures or verbal communication to ensure safety at work		
8	If necessary, explain the purpose of his work with expert arguments		
KNOWLEDGE & UNDERSTANDING-you need to know & understand:			
3	Knows the dangers that can arise when felling a tree		
3	Knows the regulations and guidelines regarding the height of stumps		
3	Knows the process of felling a tree that is thicker than the length of the processor head bar		

3	Knows the difference between wood for wood and wood for pulp and wood		
3	Knows the consequences of damage to standing trees with forestry machinery		
3	Knows how to cross transverse ditches and riverbeds		
5	Commercial works: understands costs of fuel, labour and materials etc		
8	Knows the rules of teamwork		
8	Knows professional terminology		

NOS patterns were analysed. This meant combining the two related UK standards for reflection with the single standards for forest machine operations of Czech Republic and Slovenia (Tabs. 13, 14 & 15).

Table. 13 – Mapping document 3a-Thematic analysis of UK Forestry Harvester standards (Robb 2019)

Theme	Harvester & Base Machine
Country	United Kingdom
Standard Name and Code	Fell trees mechanically (code: LANTw38) & Prepare and operate a base machine with attachments (code: LANTw79)
Responsible authority	Skills Development Scotland
Validity of the standard	From February 2017
Qualification level	EQF (undefined) although promoted by providers at level 3
Name/Scope of professional competence level	Harvester 1-How to use harvesting and base machinery with attachments 2-Machinery settings 3-Felling trees 4-Site preparation

	Base machine (undefined)
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Table. 14 – Mapping document 3b-Thematic analysis of CZ Forestry Harvester standards (Robb 2019)

Theme	Harvester & Forwarder
Country	Czech Republic
Standard Name and Code	Timber harvesting with logging and transport machines (code: 41-091-M)
Responsible authority	Ministry of Agriculture
Validity of the standard	From 19.6.2020
Qualification level	EQF 4
Name/Scope of professional competence level	1-Adherence to and application of health and safety principles in the operation of harvesters and transportation machinery 2-Control and operation of multi-operational harvesting machines 3-Control and operation of forwarders 4-Adjustment, care and maintenance of harvester, hydraulic arms and felling heads

Table. 15 – Mapping document 3c-Thematic analysis of SI Forestry Harvester standards (Robb 2019)

Theme	Harvester & Forwarder
Country	Slovenia
Standard Name and Code	Operator / operator of machines for machine felling and export of wood with articulated semi-trailer (code: 78281880)
Responsible authority	Ministry of Labour, Family, Social Affairs and Equal Opportunities
Validity of the standard	From 22.05.2017
Qualification level	EQF 4

Name/Scope of professional competence level	1-Analysis, planning and organization of work 2-Preparation of work or workplace 3-Operational work 4-Administrative work 5-Commercial works 6-Quality Assurance 7-Maintenance and repairs 8-Communication 9-Protecting health and the environment
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A holistic approach to forest machine operations was adopted in Czech Republic and Slovenia whereas in the UK, instead of containing the harvester and forwarder NOS in one standard, these consisted of three separate standards. Although the thematic naming of the seventeen competence levels was varied, a common structure to the forestry harvester NOS could be extrapolated and conceptualised (Fig. 16).

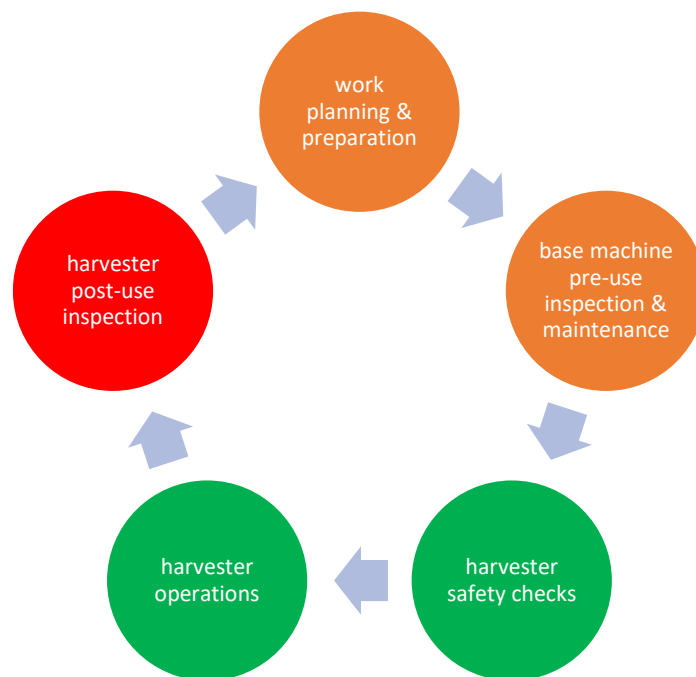


Figure. 16 – Thematic analysis results of forestry harvester competence levels (Robb 2019)

Further interpretative analysis of the NOS was used to group and classify outcomes under the harmonised competence levels. Due to the large amount

of overlap between the outcome criteria, an important aspect here was the need to avoid unnecessary repetition of outcomes and the recognition of common safety critical criteria. This was achieved along with merging tables 11 and 12 with a minor realignment of the numerical order into the new Pan-European competence levels (Tab. 16).

Table. 16 – Mapping document 4-Harmonised European Forestry Harvester standards (Robb 2019)

Outcome	European Occupational Standards-Forestry Harvester
PERFORMANCE CRITERIA-you must be able to:	
Competence Level 1-Work Planning & Preparation	
1	Assess the risks associated with the site and the proposed work & identify basic forest tree species/read forestry maps/plans as applicable
2	Select and implement appropriate working method in accordance with the assessed risks
3	Select and use appropriate personal protective equipment (PPE) for the work
4	Maintain the security of machinery and equipment on site
5	Select, prepare and use the appropriate tools and equipment, safely and effectively
Competence Level 2-Base machine pre-use inspection & maintenance	
6	Make sure the attachment is in transport position when not in use, in accordance with specifications
7	Carry out routine operator maintenance, pre-start checks (includes safety guards) and the setting of the machine for use in felling trees (includes operator environment) including replacement of bar, chain & hydraulic hose as necessary & fuel/lubricant checks
Competence Level 3-Harvester safety checks	
8	Confirm all equipment has been checked and is fit for purpose including attachments (also level 4)
9	Check the immediate work area for hazards and obstacles before moving off (use appropriate warning signals at all times)

Competence Level 4-Harvester operations	
10	Select, prepare and maintain access and egress routes in a serviceable condition
11	Operate and manoeuvre the base machine safely, and in a way that is consistent with the type of vehicle, attachment and operation
12	Modify operating procedures to take into account any changes in weather, ground conditions, types of terrain and site restraints
13	Utilise additional safeguards when reversing or manoeuvring and comply with safety distances adjacent to roads and tracks or where others are working
14	Maintain the efficiency of base machine and attachment performance through the appropriate handling and maintenance
15	Deal with any hazards and obstacles encountered during the operation, in accordance with standard practice & communicate with co-workers with gestures or verbal communication to ensure safety at work
16	Monitor the operation of the attachment and recognise any faults
17	Ensure that damage to the worksite, any remaining standing trees, tracks, roads, drains and the wider environment is kept within specified limits
18	Identify the trees to fell and position the machine to fell trees in the required direction, for both clear-fell and thinning operations
19	Select a felling method that is relevant to the tree size and condition and to the given specification & ensure nobody is in the danger area
20	Fell trees mechanically, safely and effectively according to the specification
21	Position timber appropriately for subsequent processing
22	Meet given specifications for stump height (and stump treatment, where applicable)

Competence Level 5-Harvester post-use inspection	
23	Stop and isolate the base machine on completion of the activity, in line with standard practice
24	Remove the attachment, where appropriate, in accordance with specifications
25	Leave the base machine and attachment secure after use and in a suitable condition for future use
26	Ensure the site is left in a condition which meets environmental requirements, in accordance with the specification (also level 1)
27	Maintain the health and safety of yourself and others at all times, in accordance with relevant legislation (all levels)
KNOWLEDGE & UNDERSTANDING-you need to know & understand:	
Competence Level 1-Work Planning & Preparation	
1	How to identify hazards and assess risks
2	How to interpret risk assessments
3	The types of hazards and obstacles that may be encountered and how these should be dealt with
4	The selection, use and care of personal protective equipment (PPE)
5	Site procedures including communication systems and emergency planning & knows the rules of teamwork
6	Why it is important to maintain the safety and security of equipment and vehicles on site & understands costs of fuel, labour and materials etc
7	The current guidelines on machinery operation, risk zones and safety clearances from overhead electricity conductors, working at height and what to do in the event of contact with power lines
8	How to set up and use fuelling and maintenance areas where appropriate
9	The types of tools and equipment required and how to maintain and use these safely and effectively in line with the manufacturer's recommendations
Competence Level 2-Base machine pre-use inspection & maintenance	

10	The procedure for loading and unloading base machine and attachments from transporter where appropriate
11	The legal requirements for checking equipment
12	How to mount and dismount the base machine safely
13	How to start and stop the base machine including the correct procedures for jump starting
14	The types of attachments that are safe for use with the base machine and how they are fitted
15	The conditions that should be taken into account when considering the use of attachments
16	How to set or calibrate attachments, where required
17	How to check, maintain and sharpen the felling mechanism and the procedures to ensure the operator is safeguarded
18	The requirements for machine and operator protection structures
19	Correct fitting of traction aids and advantages and disadvantages of using them
20	How to preserve the efficiency of the base machine and attachment through routine operator checks and basic maintenance in line with specifications
21	How to carry out basic repairs and where to refer problems with the base machine or attachment & keep records including machine hours etc
22	The controls and electrical services on the base machine and the attachment
23	The faults, defects and wearing of parts that occur with attachments used for treework operations
24	The function of all operating controls for base unit, machine boom/attachment and felling head, the fire control system, including on board computer system where fitted
Competence Level 3-Harvester safety checks	
25	The required pre-start checks and adjustments for the base machine and attachments, and the reasons why these must be carried out
Competence Level 4-Harvester operations	

26	The appropriate use of warning signals and indicators
27	The ways in which the base machine should be operated and manoeuvred, and how different weather, ground conditions, types of terrain and the work being carried out must be taken into account
28	The various driving characteristics and techniques for driving a base machine with different wheel configurations
29	How to operate and use attachments safely including the counterbalance
30	How to put the attachment into transport position and work position, where appropriate
31	The capabilities of the base machine and the factors that may affect its efficiency
32	How to monitor the operation of the attachment and the actions that might be required
33	How to recognise the trees to be felled mechanically and how to select the size and species to meet the required specification & check the accuracy of the required assortment lengths and the correctness of assortment diameters
34	The implications of felling double, shattered or other malformed trees, large trees (over felling bar capacity), trees with decay & the dangers that can arise when felling a tree
35	How to deal with windblown trees with over-size, buried or inaccessible butts & knows the difference between wood for wood and wood for pulp and wood
36	The legal constraints and requirements for felling trees mechanically & guidelines regarding the height of stumps
37	The capabilities and limitations of the machinery used to include its slope limitations & how to cross transverse ditches and riverbeds
38	The potential impact of your work on the environment and how this can be minimised including the consequences of damage to standing trees with forestry machinery
Competence Level 5-Harvester post-use inspection	

39	The reasons why the base machine and attachment should be left in a suitable condition for future use
40	Your responsibilities under relevant environmental, health and safety legislation and codes of practice &, if necessary, explain the purpose of their work with expert arguments and understand professional terminology

5.2.3 Data set 2: Forestry forwarder analysis

The next phase required a similar mapping process as before but this time focused on NOS for forestry forwarders (Annex A: Tab. 17). The UK standard LANTW79-Prepare and operate a base machine with attachments has been integrated, for practical purposes, within the mapped standards to avoid unnecessary duplication of content (highlighted text and coded (b)). A few criteria that appeared to be omitted but present within the harvester standard and relevant were included (h/b). The main documents analysed and mapped here were as follows:

- NOS LANTw79: Prepare and operate a base machine with attachments (2017)
- NOS LANTw47: Operate grapple loader (2017)
- NOS NSK 41-091-M: Timber harvesting with logging and transport machines (2020)
- NOS NPK-78281880: Operator / operator of machines for machine felling and export of wood with articulated semi-trailer (2017)

The data obtained from the forestry forwarder standards evaluation was compared to measure the level of similarity (Figs. 17 & 18).

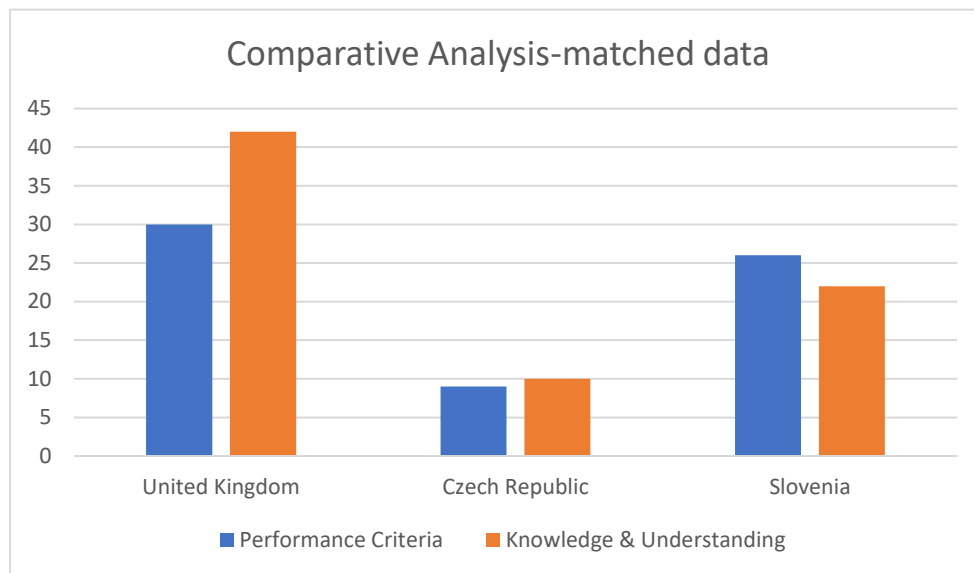


Figure. 17 – Evaluation of similarity between matched NOS learning outcomes in different forestry forwarder standards (Robb 2019)

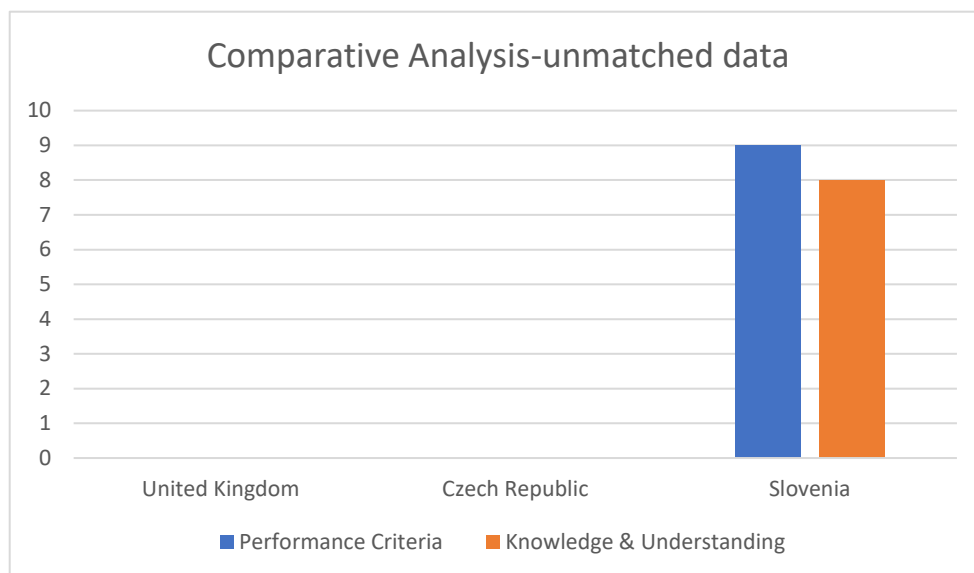


Figure. 18 – Evaluation of similarity between unmatched NOS learning outcomes in different forestry forwarder standards (Robb 2019)

Additional learning outcomes beyond the scope of the benchmark UK standards were identified (Annex A: Tab. 18).

NOS patterns were analysed. This meant combining the two related UK standards for reflection with the single standards for forest machine operations of Czech Republic and Slovenia (Annex A: Tabs. 19, 20 & 21).

As previously described, a holistic approach to forest machine operations was adopted in Czech Republic and Slovenia whereas in the UK, instead of containing the harvester and forwarder NOS in one standard, these consisted of three separate standards. Although the thematic naming of the thirteen competence levels was varied and the scope of the UK's was undefined, a common structure to the forestry forwarder NOS (similar to the harvester) could be extrapolated and conceptualised (Fig. 19).

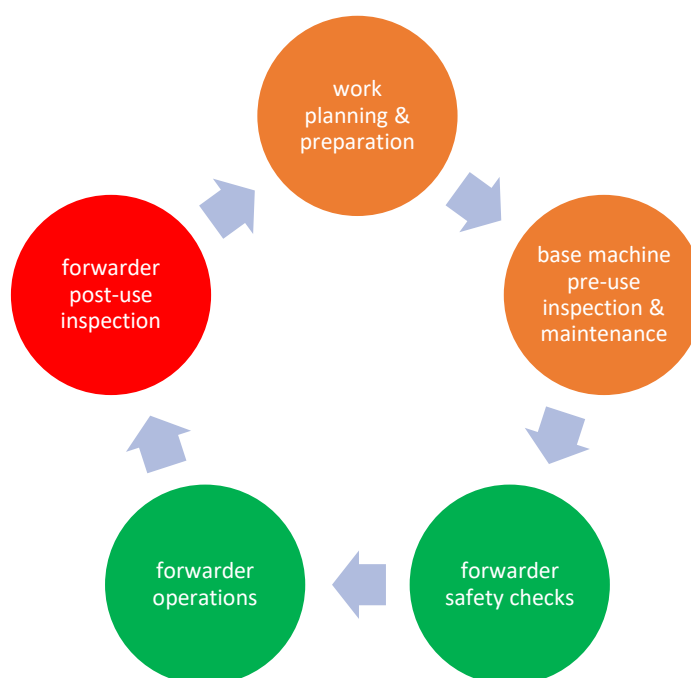


Figure. 19 – Thematic analysis results of forestry forwarder competence levels (Robb 2019)

Further interpretative analysis of the NOS was used to group and classify outcomes under the harmonised competence levels. This task was undertaken similar to the harmonisation of the harvester standards previously completed. New Pan-European competence levels were developed (Annex A: Tab. 22).

5.2.4 Data set 3: Forestry tractor/skidder analysis

The final phase in the mapping process focused on NOS for forestry tractor/skidders (Annex B: Tab. 23). The UK standard LANTW79-Prepare and operate a base machine with attachments has been integrated, for practical purposes, within the mapped standards to avoid unnecessary duplication of content highlighted text and coded (b), LANTw43 coded (m),

LANTw42 coded (w) and LANTw46 coded (s). A few criteria that appeared to be omitted but present within the harvester standard and relevant were included (h). The main documents analysed and mapped here were as follows:

- NOS LANTw79: Prepare and operate a base machine with attachments (2017)
- NOS LANTw43: Extract wood and wood products using motorised machines (2017)
- NOS LANTw42: Extract timber using mechanically powered winches (2017)
- NOS LANTw46: Attach and detach timber onto a skidder (2017)
- NOS NSK 41-091-M: Timber harvesting with logging and transport machines (2020)
- NOS NSK 41-023-H: Logging operations part d (2008) coded (l)
- NOS NPK-78281880: Operator / operator of machines for machine felling and export of wood with articulated semi-trailer (2017)
- NOS NPK-6718282011: Forestry tractor (2018) coded (f)

The data obtained from the forestry tractor/skidder standards evaluation was compared to measure the level of similarity (Figs. 20 & 21).

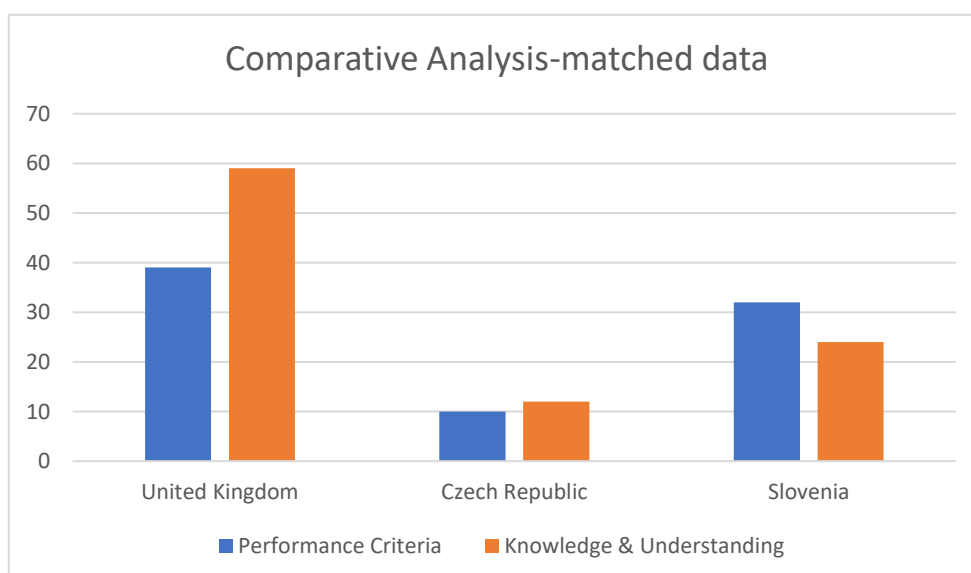


Figure. 20 – Evaluation of similarity between matched NOS learning outcomes in different forestry tractor/skidder standards (Robb 2019)

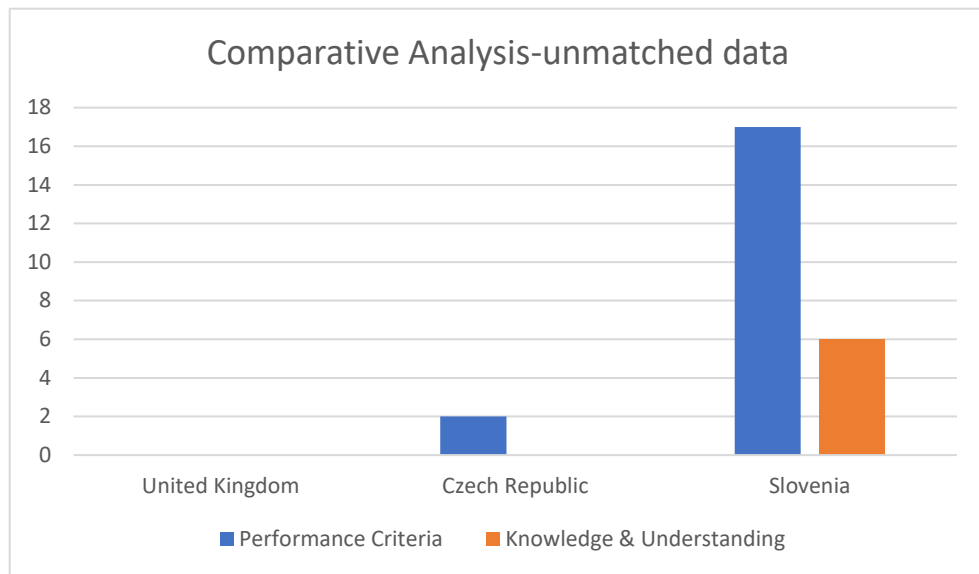


Figure. 21 – Evaluation of similarity between unmatched NOS learning outcomes in different forestry tractor/skidder standards (Robb 2019)

Additional learning outcomes beyond the scope of the benchmark UK standards were identified (Annex B: Tab. 24). NOS patterns were analysed. This meant combining all the related NOS of each country (Annex B: Tabs. 25, 26 & 27). As found with the two previous data sets a common structure to the forestry tractor/skidder NOS could be extrapolated and conceptualised (Fig. 22).

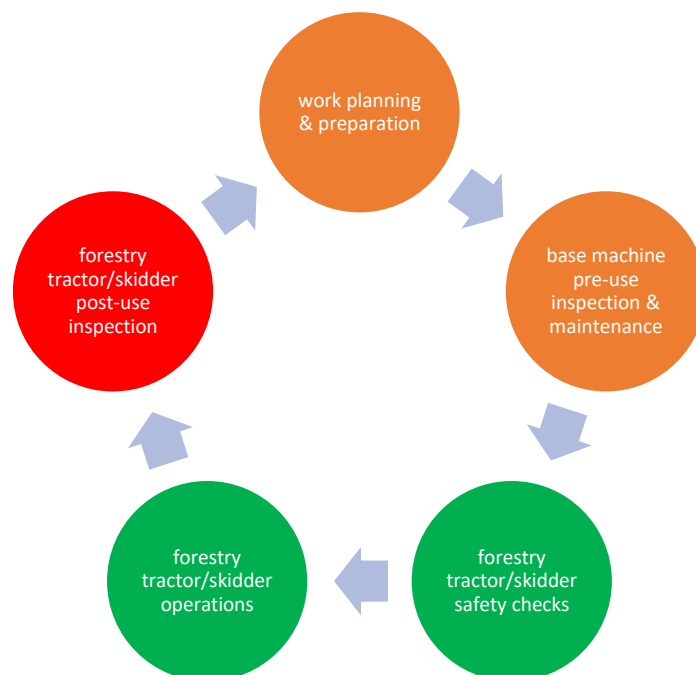


Figure. 22 – Thematic analysis results of forestry Tractor/Skidder competence levels (Robb 2019)

Further interpretative analysis of the NOS was used to group and classify outcomes under the harmonised competence levels. This task was undertaken similar to the harmonisation of the other standards previously completed. New Pan-European competence levels were developed (Annex B: Tab. 28).

On completion of this stage, the innovative and state of the art Pan-European occupational standards for forest machine operations which were developed, provided the basis for establishment of the three envisaged EFMO qualification levels.

5.3 Thesis aims and objectives 4 and 5

Following completion of the standards the next steps required the development of the qualifications into a concrete form, as outlined in thesis aims and objectives 4 & 5 as follows:

- Establish EFMO European occupational standards that can be used to develop qualifications for use by an appropriate international awarding body. This will include completion of reviews and finalising the draft standards ready for use through engagement with training providers within Europe (**no.4**)
- Perform a pilot test of EFMO European occupational standards in practice and evaluate the feedback received. Verify with training providers the final version of the standards for use in training and assessment of EFMO candidates. Finalise, live test and place on the market (**no.5**)

5.3.1 Development of new EFMO qualifications

In conjunction with ABA an EFMO certification scheme was established. This set out the framework, guidance and rules for management and delivery of the qualifications once they were finalised. This would be delivered via an already well-established network of quality assured ABA accredited national centres. Prior to this and completion of the objectives a number of standard setting events and pilot tests were necessary.

5.3.2 Standards setting and Pilot testing

Standards setting is a continuation of the detailed technical evaluation process undertaken previously as part of the mapping and development of EFMO competency levels. It was important that before the three new Pan-European qualifications became available they were checked to ensure the new competency assessments were fit for purpose. For consistency the standards setting alongside pilot testing, was mainly undertaken in cooperation with the state forest agency in Ireland-Coillte. Essentially the pilot tests were a means to verify the standards set. As a registered ABA centre with the resources, trainer/assessors and potential forestry candidates Coillte was an appropriate choice for this role. Other ABA centres in the UK and Latvia provided additional resources and supported pilot tests also.

As well as checking the content of the standards in practice there needed to be a process of determining how many marks/score a candidate needs to achieve a pass on a competency assessment. As highlighted previously because forestry is a particularly hazardous occupation it was very important to identify safety critical criteria. Following several exploratory meetings an EFMO Expert Group was established. A standards matrix developed by myself was reviewed by the group and following feedback updated (Tab. 29).

Table. 29 – Extract from ABA Pan-European NOS Forwarder: Competence Level 1-Work Planning & Preparation (Robb 2019)

INTERNATIONAL OCCUPATIONAL STANDARDS: 2019		Diagnostic tools				Level of importance		
		Written	Practical	Oral	Other	Minor	Major	Critical
EFMO-F: Forestry Forwarder Operations								
What the forest machine operator must be able to do: (Practical Standard-max time allowed 4hrs) Pre-requisite: none (Min. age 16yrs) <u>emergency 1st aid & manual handling strongly advised</u>								
UNIT 1								
LO-1	SELECT & WEAR PERSONAL PROTECTIVE EQUIPMENT-Candidate to wear appropriate PPE & show ID:							1
1:1	Candidate photo ID & SRA signed		✓					✓ 1:1
1:2	Appropriate trousers (ex. chainsaw)		✓					✓ 1:2
1:3	Appropriate safety boots/footwear (ex. chainsaw)		✓					✓ 1:3

1:4	Safety helmet, hard hat or head protection as appropriate (uv indicator/date)	✓					✓	1:4
1:5	Eye, Face & Ear protection as appropriate (ex. oil pressure leaks paper check, chainsaw use)	✓					✓	1:5
1:6	Hand protection as appropriate (ex. oil resistant gloves)	✓					✓	1:6
1:7	Non-snag outer clothing/Jewelry/Long hair/Hi-Viz upper body	✓				✓		1:7
1:8	In date personal 1 st aid kit includes a minimum large absorbent wound dressing - on person (+ Team 1 st aid kit as appropriate)	✓					✓	1:8
1:9	Whistle/Operational Mobile or Radio (according to national standards)	✓					✓	1:9
LO-2	COMPLETE A WORKPLACE RISK ASSESSMENT & EMERGENCY PLANNING PROCEDURES -Candidate to identify hazards and risks relevant to <u>use of the forwarder & site</u> to be worked on and task, how the task will be done (method statement), how to deal with an emergency (specific 1 st aid scenario) & <u>provide</u> emergency information:							2
2:1	RISK ASSESSMENT-walk site (state min.3 <u>site</u> H/R & min.3 <u>task</u> H/R & describe method to be used when doing the task)	✓	✓				✓	2:1
2:2	EMERGENCY INFORMATION – verbal (state min. 8 key requirements-names, locations, numbers. approx. direction of travel to emergency contacts etc & how to deal with a 1 st aid scenario)		✓				✓	2:2
LO-3	MEET LEGAL & SITE ENVIRONMENTAL REQUIREMENTS IN ACCORDANCE WITH NATIONAL STANDARDS -Candidate to comment on protection of flora, wildlife, waterways etc & site-specific conditions met regarding pollution/damage/biosecurity:							3
5:1	State national legislation regards environmental protection (min.1 each): A-Personal/Public B-Plants C-Animals		✓				✓	5:1
5:2	State possible causes of damage/pollution in the work site, how to prevent & deal with it (min.3 examples)		✓				✓	5:2
5:3	State the importance of bio-security & how to prevent cross-contamination (min.1 example)		✓				✓	5:3
UNIT 2								

The competency levels were designed to accommodate a range of diagnostic tools for assessment including identification of minor, major or safety critical components. This format was consistent for all three EFMO qualifications. A record of assessment based on the standards was then developed by myself to enable the pilot test phase to begin. This was reviewed by the expert group and updated taking into consideration their comments (Tab. 30).

Table. 30 – Extract from ABA Pan-European NOS Forwarder: Record of Assessment (Robb 2019)

RECORD OF ASSESSMENT (ROA): Assessment Criteria EFMO-F		Feedback comments to candidate & Result v/x	
FORESTRY FORWARDER OPERATIONS Pre-requisites: None but <u>emergency 1st aid & manual handling strongly advised</u> Max. Time Allowed – 4hr			
1. (UNIT 1) SELECT & WEAR PERSONAL PROTECTIVE EQUIPMENT (Assessor to inspect/validate) Assessor to candidate max ratio 1:1 on an independent 3rd party capacity only			
Candidate to inspect, wear appropriate PPE & show photo ID:		CANDIDATE	
1. Candidate photo ID & SRA signed	C	LOCATION	
		MACHINE MODEL/TYPE Wheeled ☐ Tracked ☐	
2. Appropriate trousers (ex. chainsaw)	C	CAB ☐ NO CAB ☐	
3. Appropriate safety boots/footwear (ex. chainsaw)	C	DATE/...../.....	Start Time
4. Safety helmet, hard hat or head protection as appropriate (uv indicator/date)	C		
5. Eye, Face & Ear protection as appropriate (ex. oil pressure leaks paper check, chainsaw use)	C		
6. and protection as appropriate (ex. oil resistant gloves)	C		
7. Non-snag outer clothing/Jewelry/Long hair/Reflective Hi-Viz upper body etc		PPE checks: • Candidates should be aware of equipment required for test (owner's manual available) • Candidate assists assessor to complete SRA before start of test • Gloves/spill kit/barrier cream • Fire extinguisher	
8. In date personal 1 st aid kit includes a minimum large absorbent wound dressing - on person (+ Team 1 st aid kit as appropriate - on site)	C		

9. Whistle/Operational Mobile or Radio (according to national standards)	C	
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As seen previously (Fig. 12) pilot tests begun during October 2019. Standard setting also took place practically during these field activities, as the ABA assessors needed clarification, on how to determine and record the assessment results correctly. The assessors undertook several mock sessions with each other, in order to gain familiarity with the record of assessment, prior to the actual pilot assessments which were verified by an ABA forest machinery verifier. This process was observed and reviewed by Coillte training specialist's/health and safety officers and other members of the expert group including myself. Numerous pilot tests and standard setting events ('fine-tuning' the standards) were held until the lead assessors and verifiers were satisfied that we were ready to undertake the final phase-live tests.

Following feedback, I undertook the preparation of pre-assessment qualification guidance for candidates prior to the live tests (Tab. 31).

Table. 31 – EFMO pre-assessment guidance (Robb 2019)

Qualification Guidance	Tick
Preparatory training: Candidates are strongly advised to undertake a <u>training course</u> with an approved ABA Centre/Training Provider prior to assessment in addition to obtaining evidence of suitable <u>emergency 1st aid & manual handling</u> training as applicable. Are you suitably prepared & trained for the assessment?	
Candidate registered prior to assessment. Are you registered with an ABA Centre?	
Prerequisites met: check with ABA Centre/Training Provider. Have you met any required prerequisites?	
Proof of identity evidenced: photo drivers' licence, passport, national photo ID or govt authorised document + photo. Do you have the correct photo ID to show the assessor?	
Operators' manual. Spill kit, Fire extinguisher, Maintenance tools etc. Are they available for the assessment?	
Do you have the appropriate personal protective equipment (PPE) & 1st aid available for the assessment?	
Qualification guidance: ABA issue detailed qualification guidance for each award which includes access requirements, quality assurance and any other information deemed necessary to be understood prior to an assessment taking place. Have you read & understood the Qualification Guide?	

Record of assessment (ROA): C=Critical Faults & termination of assessment others count towards assessment of overall grade achieved. Additionally, the Record of Assessment includes additional qualification guidance that should be read & understood. Have you read & understood the ROA?	
Machine familiarity: For example, knowledge of working at heights in relation to changing a hose on outer/main boom and replacing a chain/bar on a harvester head with the head parked safely on the ground. Are you familiar with the machine?	
Machine suitability: Is the machine 'fit for purpose' i.e., is it in good working condition, meet the relevant regulations & requirements such as operator protection, fire safety etc. Is the machine suitable for assessment?	
Notes:	
Best Practice Guidance	
Pre-assessment: review local, national & relevant European legislation/regulations & other sources of safe operation guidance & OSH guidance particularly the operator's manual. Have you reviewed relevant best practice guidance?	
Source Examples only-Ireland: IFSG 503 https://www.hsa.ie/eng/Publications_and_Forms/Publications/Agriculture_and_Forestry/Extraction_by_Forwarder - IFSG 503.pdf	
IFSG 603 https://www.hsa.ie/eng/Publications_and_Forms/Publications/Agriculture_and_Forestry/Mechanical_Harvesting - IFSG 603.pdf	
IFSG 804 https://www.hsa.ie/eng/Publications_and_Forms/Publications/Agriculture_and_Forestry/Electricity_at_Work - IFSG 804.pdf	
Forwarder Manual-John Deere https://partscatalog.deere.com/jdrc/navigation/equipment/3976120	
Biosecurity-UK Forestry Commission https://www.forestryengland.uk/sites/default/files/pdf/Biosecurity%20Policy.pdf	
Notes:	

As of March 2022, ABA has certified 65 successful machine operators in Ireland. They have been independently assessed (the assessor was not involved in their training). The certification is valid for five years and is available in digital form via the ABA website (Annex C: Fig. 23). The certification is also graded and the qualification reviewed regularly.

6. Discussion

The core aim of the research was to investigate forest machine qualifications, analyse the data, develop quality Pan-European qualifications and implement them into the market place. It was found that in Czech Republic a certificate to operate a hydraulic crane and a T class driving licence which covers tractor operations, are compulsory requirements to work with forest machines (Personal communication, NOS NSK41-091-M). This imposes limitations to working on a Pan-European context as the licence and certificate are not harmonised therefore only valid for use in Czech Republic. It is a similar situation whereby the Slovenian F class driving licence is required (NOS NPK-6718282011). This also restricted the ability to thoroughly map base machine standards within mapping document no.1. This was identified at the literature review stage as results 4 and 5 (Tab. 10). Base machine is a separate standard in the UK, therefore a practical solution was to integrate it throughout the three qualifications levels developed (harvester, forwarder and skidder) which ensured the operator, no matter what their background qualifications were, could be determined competent.

There was clearly an increased machine operator demand ((Fobia 2019) (European Commission 2021b)) and increased risks of injury in the sector ((Klun and Medved, 2007) (Axelsson 1998) (Campu 2021) (M Lee et al 2017)), including prior to this study, the absence of Pan-European forestry machine qualifications. These were the findings of the literature review namely results 1,2,3 and 4 (Tab. 10). Result no.10 of the literature review supported the use of ILO resources or those such as the EQF in the development of EFMO qualifications (Tab. 10). Although care had to be taken, as it was found that the EQF was more suited to academic attainment (Lester 2015). Additionally, it was found that UNESCO (2021) supports the delivery of vocational training externally of state, via non-profits for example, due to the fact that they are more closely aligned to trade and the world of work. The certification body ABA International is a non-profit organisation therefore well suited to implement the results of this research. As found stated by OSH increased engagement with partner countries and international organisations can raise OSH standards worldwide (European

Commission 2021d). Literature review results 6-11, outlined the multiple benefits to be gained, from implementation of the EFMO qualifications for employees, employers and stakeholders across Europe and internationally (Tab. 10).

The results of the literature review also found employers must ensure that workers given the task of using work equipment, receive adequate training by law (EU 2009) and independent assessment (separated from training) of a forest machine operators' competency is the recommended way of proving this (FISA 2013). These two components were reflected in the development of the EFMO qualifications. Indeed, as were the numerous benefits associated with adequate training and refresher training requirements ((Beck 2020) Nieuwenhuis and Lyons (2002) (Axelsson 1998) (Rutkowska & Adamowicz 2018) (Aalmo and Talbot 2014) (Cedefop 2015)).

During the mapping process between the selected countries some interesting results became evident. In relation to the comparative analysis of harvester standards (Figs. 14 & 15), the UK and Czech data was distinctly skewed towards a larger requirement for knowledge criteria, from the forest machine operator, which was almost twice the level of their practical performance criteria requirements. It is possible that this is an indication that the EQF structure was more suited to academic attainment as stated by Lester (2015) and therefore has limited effectiveness in the more vocational environment. This would also explain why the Slovenian data appeared to be a more balanced reflection of skills and knowledge requirements. This is because the Slovenian standards, have a distinctly more work/task-based structure whereby performance and knowledge criteria are not separated but are combined, as key works within a distinct field of work. Further research is needed here. Gaps in skills and knowledge within the UK standards were also identified in the analysis (Tab. 12) but these were integrated within the final EFMO NOS.

Competency levels for harvester operations varied somewhat but through thematic analysis five core areas applicable to all the sample countries was identified (Fig. 16). After harmonisation and realignment an EFMO forestry harvester standard was developed (Tab. 16). Interestingly, competency

levels in Czech Republic are referred to as professional competency levels and in Slovenia, the complete standards are referred to as professional standards.

The mapping process with data set 2 for forestry forwarders found a similar pattern as with harvesters. Although for the UK and Czech data was skewed towards knowledge criteria it was less than before. Gaps in skills and knowledge within the UK and Czech standards were also identified in the analysis which were integrated within the final forestry forwarder standard (Tab. 22). Following thematic analysis competency levels were found to be same as the forestry harvester levels.

The mapping process with data set 3 for forestry tractors/skidder found a similar pattern as with forwarders. Although for the UK and Czech data was skewed towards knowledge criteria it was less than before. Gaps in skills and knowledge within the UK standards were also identified in the analysis which were integrated within the final forestry tractor/skidder standard (Tab. 28). Interestingly the skills/knowledge gap was larger here between Slovenia and the others, which was potentially due to the effect of local conditions. Due to the more mountainous environment and tradition of skidder extraction, the Slovenian standards were more advanced. This was seen as a benefit to the new EFMO qualification. Following thematic analysis competency levels were found to be same as the forestry harvester and forwarder levels.

Subsequent development of the EFMO NOS (Tab. 29) included the integration of diagnostic tools and levels of complexity from minor to critical, which would then inform a grading system to be used by assessors, on a form known as the record of assessment (Tab. 30). Once completed these were then field tested in the next phase.

Standard setting and pilot tests (which were integrated) were undertaken primarily in Ireland but also elsewhere, in support of the evaluation process. Fourteen pilot tests to date have been undertaken (Tab. 9). This was a key part of the process whereby the products could be checked, tested, reviewed and updated, in practice in the field. Standard setting would be undertaken

numerous times within each pilot test. As stated by Ben-David (2000), increased standard-setting enhances high-quality testing and raises confidence in the testing of professional competencies. Pilot test parameters in practice were a minimum of four with at least the last two assessment durations remaining consistent. Pilot testing was completed earliest for forestry harvesters and therefore became the first EFMO qualification to be live tested. It took seven pilot tests to be concluded before the forestry forwarder could be live tested. Although the forestry tractor/skidder EFMO qualification appears almost ready, at least one further pilot test will be undertaken before live tests commence during 2022. Qualification guidance (Fig. 31) was prepared as a result of the pilot tests and training support materials which are due for completion by the end of April 2022.

ABA has certified 65 successful machine operators in Ireland. They have been independently assessed (the assessor was not involved in their training). The certification is valid for five years and is available in digital form via the ABA website. This has proven to be a very useful verification tool for employers or potential employers, where a candidate has provided their unique identification number and date of birth. The certification is also graded from A+ to F (fail) therefore it is not based on a purely pass/fail basis. The qualifications are reviewed on a regular basis which is more frequent than the five yearly NOS reviews by many state authorities.

7. Conclusion

The UK, Czech and Slovenian systems were researched in the case of EFMO, as it was hypothesised, to encourage quality standards of training to ensure candidates are able to pass an independent assessment. In theory and in practice this hypothesis has been proven, as the structure and content of the EFMO qualifications as described within this study, are enhanced versions of the national ones and have been concretely implemented into the market place.

The individual objectives include as follows:

1. Research the issue of European Forest Machine Operator Qualification Standards and present the findings. Undertake a literature review to ascertain the current position in this field.
 - Final comment: Achieved and presented demonstrating a need for high quality European wide qualifications.
2. Develop European Forest Machine Operator (EFMO) learning outcomes based on performance criteria & underpinning knowledge in the operation of tractors (skidders), forwarders and harvesters in forestry. This includes an investigation of the required job tasks and level of knowledge to work safely. The learning outcomes will be mapped against any previous relevant findings.
 - Final comment: Has been completed and enhanced versions developed from previous findings and incorporated into new qualifications.
3. Align the new EFMO learning outcomes with the European Qualification Framework (EQF). Investigate the EQF processes and how they can be applied to provide an indication of the appropriate levels of the new EFMO learning outcomes.
 - Final comment: Has been completed and qualifications rated at EQF level 4.
4. Establish EFMO European occupational standards that can be used to develop qualifications for use by an appropriate international awarding body. This will include completion of reviews and finalising the draft standards ready for use through engagement with training providers within Europe.

- Final comment: Has been completed and in use by providers through the ABA International network.

5. Perform a pilot test of EFMO European occupational standards in practice and evaluate the feedback received. Verify with training providers the final version of the standards for use in training and assessment of EFMO candidates. Finalise, live test and place on the market.

- Final comment: Many tests were successfully undertaken including live tests with two of the three products in use on the market.

I fully endorse and recommend the new EFMO qualifications for use in operational forestry practice for the following key reasons:

- Due to high standards and rigorous independent assessment, they can help to improve forestry safety standards and safety culture
- They can improve the recognition of forest machine operators' qualifications and effectively reduce any potential barriers to employment & transnational movement of labour
- They can support the overall international competitiveness of the European forestry labour force on a global level

8. Souhrn

Poslední verze harmonizovaných kvalifikací EFMO pro provoz harvestorů, vyvážecích traktorů a lesních traktorů byly úspěšně dokončeny a uvedeny na trh.

Vypracování Cíle 1 jasně ukázalo potřebu vysoce kvalitních evropských kvalifikací. Použité metody, jako jsou rozhovory a literární rešerše, poukázaly na problémy v této oblasti, jako míra nehodovosti v tomto sektoru, překážky v zaměstnání, neuznávání různých národních kvalifikací atd. Zároveň byla identifikována rostoucí poptávka po kvalifikovaných operátorech lesnické techniky. Byly také nastoleny otázky týkající se vhodnosti státem řízené profesní certifikace včetně mechanismů jejího udělování. Zda jsou tyto postupy vhodné pro daný účel v moderním a technologicky náročném dynamickém pracovním prostředí, si zaslouží další výzkum.

Cíle 2 a 3 zahrnovaly kvalitativní metodologii mapování standardů a analýzy dat. Tento přístup, který byl proveden ve třech fázích, byl velmi užitečný při identifikaci relevance a podobnosti dat. Například chyby, duplikace a mezery mezi standardy různých zemí byly v tomto procesu opraveny, což bylo výhodou při dosažení cílů této práce. Úspěšně byly aplikovány metody obsahové, srovnávací, tematické a interpretační analýzy. Získané soubory dat zdůrazňovaly nesrovnalosti v množství požadovaných znalostí ve srovnání s dovednostmi, což bylo pozorováno častěji u norem Spojeného království, zatímco ve srovnání se slovinskými normami se zdálo, že mají vyváženější přístup při zachování důrazu na dovednosti. Získané výsledky naznačují potřebu dalšího výzkumu procesu certifikace a metodologie aplikované ve Slovinsku.

Příspěvek této práce v oblasti lesnictví se již úspěšně promítl do splnění cílů 4 a 5. Byly vyvinuty nové detailní evropské standardy a kvalifikace, které byly po četných pilotních testech implementovány na trh. Kvalifikace jsou nyní uváděny do praxe také v jiných zemích mimo původní testovací prostředí Irsko. Nové, časově limitované úrovně kompetencí, které byly vyvinuty, poskytují realističtější a užitečnější měřítko dovedností

pracovníka, než jen zjednodušený systém prospěl/neprospěl. Certifikační orgán ABA International poskytuje mnoho dalších výhod jako jsou např. neustálá podpora a další rozvoj standardů dle potřeb průmyslu. Jinými slovy, byla zajištěna dlouhodobá udržitelnost výsledků.

Kvalifikace EFMO budou v budoucnu dále rozvíjeny s využitím dalšího výzkumu a širších konzultací s mnoha zainteresovanými stranami z celé Evropy. S touto skutečností úzce souvisí přínosy pro tento sektor jako jsou zvýšení uznávání kvalifikací a profesionality operátorů. V lesnictví jsou naléhavě potřebné dva klíčové faktory, které tato studie ukázala. Slovinské standardy uvedené v této studii jsou skutečně nazvané „Profesní standardy“, a proč ne, koneckonců lesnictví je povolání a zaslouží si být jako takové uznáno.

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APPENDIX A: Data Set 2

Table. 17 – Mapping document 5-UK, CZ and SI Forestry Forwarder standards evaluation (Robb 2019)

UK Outcome	European Occupational Standards- Forestry Forwarder (UK)	CZ Outcome	SI Outcome
PERFORMANCE CRITERIA-you must be able to:			
1	Assess the risks associated with the site and the proposed work	1	1
2	Select and implement appropriate working method in accordance with the assessed risks	1,2	1
3	Select and use appropriate personal protective equipment (PPE) for the work		2
4	Maintain the security of machinery and equipment on site		2
5	Select, prepare and use the appropriate tools and equipment, safely and effectively		2
6(b/h)	Confirm all equipment has been checked and is fit for purpose (including attachments)		2
7(b/h)	Carry out routine operator maintenance, pre-start checks (includes safety guards) and the setting of the machine for use (includes operator environment) including replacement of hydraulic hose as necessary & fuel/lubricant checks	4	2,3,7
8(b)	Check the immediate work area for hazards and obstacles before moving off (use appropriate warning signals at all times)		3

9	Select, prepare and maintain access and egress routes in a serviceable condition		3
10(b)	Operate and manoeuvre the base machine safely, and in a way that is consistent with the type of vehicle, attachment and operation		3
11(b)	Make sure the attachment is in transport position when not in use, in accordance with specifications		3
12(b)	Modify operating procedures to take into account any changes in weather, ground conditions, types of terrain and site restraints		1,3
13(b)	Utilise additional safeguards when reversing or manoeuvring and comply with safety distances adjacent to roads and tracks or where others are working		3
14(b)	Maintain the efficiency of base machine and attachment performance through the appropriate handling and maintenance	1	3
15(b)	Deal with any hazards and obstacles encountered during the operation, in accordance with standard practice		3
16(b)	Monitor the operation of the attachment and recognise any faults		3
17	Ensure that damage to the worksite, any remaining standing trees, tracks, roads, drains and the wider environment is kept within specified limits		3

18	Identify the timber to load and position the base unit to operate the grapple loader	3	3
19	Select the timber or other products of a size and weight compatible with the capacity of the boom and grapple	3	3
20	Use the grapple loader safely, effectively and efficiently to lift, manoeuvre and load timber or other products, as applicable	3	3
21	Use the grapple loader to flush the wood products, as appropriate		3
22	Limit use of the grapple to avoid damage to products, as applicable, when loading and unloading		3
23	Unload products safely, effectively and efficiently onto segregated stacks	3	3
24	Grade, separate and accumulate products to facilitate their subsequent handling, processing or uplift for onward dispatch	3	3
25	Meet the relevant legislative and environmental requirements relevant to operating the machine, as well as those of your organisation		
26(b)	Stop and isolate the base machine on completion of the activity, in line with standard practice		
27(b)	Remove the attachment, where appropriate, in accordance with specifications		
28(b)	Leave the base machine and attachment secure after use and in a suitable condition for future use		

29	Ensure the site is left in a condition which meets environmental requirements, in accordance with the specification		9
30	Maintain the health and safety of yourself and others at all times, in accordance with relevant legislation		9
KNOWLEDGE & UNDERSTANDING-you need to know & understand:			
1	How to identify hazards and assess risks	1	9
2	How to interpret risk assessments		5
3	The selection, use and care of personal protective equipment (PPE)		9
4	Site procedures including communication systems and emergency planning		8
5	The types of tools and equipment required and how to maintain and use these safely and effectively in line with the manufacturer's recommendations	1,4	7
6	The legal requirements for checking equipment		3
7	Why it is important to maintain the safety and security of equipment and vehicles on site		2
8	The current guidelines on machinery operation, risk zones and safety clearances from overhead electricity conductors, working at height and what to do in the event of contact with power lines		1,3
9(h/b)	The function of all operating controls for base unit, machine boom/ attachment etc	2,3,4	2,3

10	The capabilities and limitations of the grapple loader when loading, in relation to maximum safe working load		3
11(b)	The required pre-start checks and adjustments for the base machine and attachments, and the reasons why these must be carried out	1	2
12(b)	The requirements for machine and operator protection structures		
13(b)	The procedure for loading and unloading base machine and attachments from transporter where appropriate	1	
14(b)	How to mount and dismount the base machine safely		
15(b)	How to start and stop the base machine including the correct procedures for jump starting		
16(b)	The types of attachments that are safe for use with the base machine and how they are fitted		
17(b)	The conditions that should be taken into account when considering the use of attachments		
18(b)	How to set or calibrate attachments, where required	4	
19(b)	The appropriate use of warning signals and indicators		
20(b)	The ways in which the base machine should be operated and manoeuvred, and how different weather, ground conditions, types of terrain and the work being carried out must be taken into account		1,3

21(b)	The various driving characteristics and techniques for driving a base machine with different wheel configurations		
22(b)	Correct fitting of traction aids and advantages and disadvantages of using them	2	
23(b)	How to operate and use attachments safely including the counterbalance		
24(b)	The controls and electrical services on the base machine and the attachment		
25(b)	How to put the attachment into transport position and work position, where appropriate	1	
26(b)	The capabilities of the base machine and the factors that may affect its efficiency		
27(b)	The types of hazards and obstacles that may be encountered and how these should be dealt with		1
28(b)	How to monitor the operation of the attachment and the actions that might be required		6
29(b)	The faults, defects and wearing of parts that occur with attachments used for treework operations		
30(b)	How to preserve the efficiency of the base machine and attachment through routine operator checks and basic maintenance in line with specifications	4	7
31(b)	How to carry out basic repairs and where to refer problems with the base machine or attachment	4	7

32(b)	The reasons why the base machine and attachment should be left in a suitable condition for future use		
33	The slope limitations on the loading and unloading operations		1,3,9
34	The methods of loading and unloading long logs, poles or tree lengths		
35	The lifting and reach capabilities and height limitations of the machinery used when unloading		
36	How to maintain the stability of the machine during unloading and driving		9
37	How to segregate and grade products to meet the required specification when unloading		3.7 (less specific)
38	The safe stacking heights, stability of stacks and signing requirements		3
39	The use of bearers where appropriate when unloading		
40(b)	How to set up and use fuelling and maintenance areas where appropriate		
41	The potential impact of your work on the environment and how this can be minimised		9
42	Your responsibilities under relevant environmental, health and safety legislation and codes of practice		9

Table. 18 – Mapping document 6-Comparative analysis of UK, CZ and SI Forestry Forwarder standards-additional findings (Robb 2019)

CZ Outcome	European Occupational Standards-Forestry Forwarder (CZ)	UK Outcome	SI Outcome
ADDITIONAL PERFORMANCE CRITERIA-you must be able to:			

	None		
SI Outcome	European Occupational Standards- Forestry Forwarder (SI)	UK Outcome	CZ Outcome
ADDITIONAL PERFORMANCE CRITERIA-you must be able to:			
1	Get acquainted with the tasks in the harvesting units and on the job site with the help of a work order, harvesting plan or contract		
1	Reads forestry maps		
2	Check the suitability and amount of fuel and lubricants in the machine		
4	Keep records: fuel/oil consumption, machine hours, faults, maintenance etc		4 (less specific)
6	Check the accuracy of the required assortment lengths and the correctness of assortment diameters		
8	Communicates with co-workers with gestures or verbal communication to ensure safety at work		
8	If necessary, explain the purpose of his work with expert arguments		
3	Perform an empty ride with an articulated semi-trailer in the correct position		
3	Removes larger branches, bark and debris that fall from the semi-trailer onto the road during unloading so as not to obstruct traffic		
KNOWLEDGE & UNDERSTANDING-you need to know & understand:			
3	Knows the regulations and guidelines regarding the height of stumps		
3	Knows the difference between wood for wood and wood for pulp and wood		
3	Knows the consequences of damage to standing trees with forestry machinery		

3	Knows how to cross transverse ditches and riverbeds		
5	Commercial works: understands costs of fuel, labour and materials etc		
8	Knows the rules of teamwork		
8	Knows professional terminology		
3	Knows the dangers of driving with a full load or at speed		

Table. 19 – Mapping document 7a-Thematic analysis of UK Forestry Forwarder standards (Robb 2019)

Theme	Forwarder & Base Machine
Country	United Kingdom
Standard Name and Code	Operate a grapple loader (code: LANTw47) & Prepare and operate a base machine with attachments (code: LANTw79)
Responsible authority	Skills Development Scotland
Validity of the standard	From February 2017
Qualification level	EQF (undefined) although promoted by providers at level 3
Name/Scope of professional competence level	Forwarder (undefined) Base machine (undefined)

Table. 20 – Mapping document 7b-Thematic analysis of CZ Forestry Harvester standards (Robb 2019)

Theme	Harvester & Forwarder
Country	Czech Republic
Standard Name and Code	Timber harvesting with logging and transport machines (code: 41-091-M)

Responsible authority	Ministry of Agriculture
Validity of the standard	From 19.6.2020
Qualification level	EQF 4
Name/Scope of professional competence level	1-Adherence to and application of health and safety principles in the operation of harvesters and transportation machinery 2-Control and operation of multi-operational harvesting machines 3-Control and operation of forwarders 4-Adjustment, care and maintenance of harvester, hydraulic arms and felling heads

Table. 21 – Mapping document 7c-Thematic analysis of SI Forestry Harvester standards (Robb 2019)

Theme	Harvester & Forwarder
Country	Slovenia
Standard Name and Code	Operator / operator of machines for machine felling and export of wood with articulated semi-trailer (code: 78281880)
Responsible authority	Ministry of Labour, Family, Social Affairs and Equal Opportunities
Validity of the standard	From 22.05.2017
Qualification level	EQF 4
Name/Scope of professional competence level	1-Analysis, planning and organization of work 2-Preparation of work or workplace 3-Operational work 4-Administrative work 5-Commercial works 6-Quality Assurance 7-Maintenance and repairs 8-Communication 9-Protecting health and the environment

Table. 22 – Mapping document 8-Harmonised European Forestry Forwarder standards (Robb 2019)

Outcome	European Occupational Standards-Forestry Forwarder
PERFORMANCE CRITERIA-you must be able to:	
Competence Level 1-Work Planning & Preparation	
1	Assess the risks associated with the site and the proposed work & identify basic forest tree species/read forestry maps/plans as applicable
2	Select and implement appropriate working method in accordance with the assessed risks
3	Select and use appropriate personal protective equipment (PPE) for the work
4	Maintain the security of machinery and equipment on site
5	Select, prepare and use the appropriate tools and equipment, safely and effectively
Competence Level 2-Base machine pre-use inspection & maintenance	
6	Make sure the attachment is in transport position when not in use, in accordance with specifications
7	Carry out routine operator maintenance, pre-start checks (includes safety guards) and the setting of the machine for use (includes operator environment) including replacement of hydraulic hose as necessary & fuel/lubricant checks
Competence Level 3-Forwarder safety checks	
8	Confirm all equipment has been checked and is fit for purpose including attachments (also level 4)
9	Check the immediate work area for hazards and obstacles before moving off (use appropriate warning signals at all times)
Competence Level 4-Forwarder operations	
10	Select, prepare and maintain access and egress routes in a serviceable condition
11	Operate and manoeuvre the base machine safely, and in a way that is consistent with the type of vehicle, attachment and operation

12	Modify operating procedures to take into account any changes in weather, ground conditions, types of terrain and site restraints
13	Utilise additional safeguards when reversing or manoeuvring and comply with safety distances adjacent to roads and tracks or where others are working
14	Maintain the efficiency of base machine and attachment performance through the appropriate handling and maintenance
15	Deal with any hazards and obstacles encountered during the operation, in accordance with standard practice & communicate with co-workers with gestures or verbal communication to ensure safety at work
16	Monitor the operation of the attachment and recognise any faults
17	Ensure that damage to the worksite, any remaining standing trees, tracks, roads, drains and the wider environment is kept within specified limits
18	Identify the timber to load and position the base unit to operate the grapple loader
19	Select the timber or other products of a size and weight compatible with the capacity of the boom and grapple & ensure nobody is in the danger area
20	Use the grapple loader safely, effectively and efficiently to lift, manoeuvre and load timber or other products, as applicable
21	Use the grapple loader to flush the wood products, as appropriate
22	Limit use of the grapple to avoid damage to products, as applicable, when loading and unloading
23	Unload products safely, effectively and efficiently onto segregated stacks
24	Grade, separate and accumulate products to facilitate their subsequent handling, processing or uplift for onward dispatch

25	Meet the relevant legislative and environmental requirements relevant to operating the machine, as well as those of your organisation
Competence Level 5-Forwarder post-use inspection	
26	Stop and isolate the base machine on completion of the activity, in line with standard practice
27	Remove the attachment, where appropriate, in accordance with specifications
28	Leave the base machine and attachment secure after use and in a suitable condition for future use
29	Ensure the site is left in a condition which meets environmental requirements, in accordance with the specification
30	Maintain the health and safety of yourself and others at all times, in accordance with relevant legislation
KNOWLEDGE & UNDERSTANDING-you need to know & understand:	
Competence Level 1-Work Planning & Preparation	
1	How to identify hazards and assess risks
2	How to interpret risk assessments
3	The types of hazards and obstacles that may be encountered and how these should be dealt with
4	The selection, use and care of personal protective equipment (PPE)
5	Site procedures including communication systems and emergency planning & knows the rules of teamwork
6	Why it is important to maintain the safety and security of equipment and vehicles on site & understands costs of fuel, labour and materials etc
7	The current guidelines on machinery operation, risk zones and safety clearances from overhead electricity conductors, working at height and what to do in the event of contact with power lines
8	How to set up and use fuelling and maintenance areas where appropriate

9	The types of tools and equipment required and how to maintain and use these safely and effectively in line with the manufacturer's recommendations
Competence Level 2-Base machine pre-use inspection & maintenance	
10	The procedure for loading and unloading base machine and attachments from transporter where appropriate
11	The legal requirements for checking equipment
12	How to mount and dismount the base machine safely
13	How to start and stop the base machine including the correct procedures for jump starting
14	The types of attachments that are safe for use with the base machine and how they are fitted
15	The conditions that should be taken into account when considering the use of attachments
16	The requirements for machine and operator protection structures
17	Correct fitting of traction aids and advantages and disadvantages of using them
18	How to preserve the efficiency of the base machine and attachment through routine operator checks and basic maintenance in line with specifications
19	How to carry out basic repairs and where to refer problems with the base machine or attachment & keep records including machine hours etc
20	The controls and electrical services on the base machine and the attachment
21	The faults, defects and wearing of parts that occur with attachments used for treework operations
22	The function of all operating controls for base unit, machine boom/attachment etc
Competence Level 3-Forwarder safety checks	
23	The required pre-start checks and adjustments for the base machine and attachments, and the reasons why these must be carried out
Competence Level 4-Forwarder operations	

24	The appropriate use of warning signals and indicators
25	The ways in which the base machine should be operated and manoeuvred, and how different weather, ground conditions, types of terrain and the work being carried out must be taken into account
26	The various driving characteristics and techniques for driving a base machine with different wheel configurations
27	How to operate and use attachments safely including the counterbalance
28	How to put the attachment into transport position and work position, where appropriate
29	The capabilities of the base machine and the factors that may affect its efficiency
30	The capabilities and limitations of the grapple loader when loading, in relation to maximum safe working load
31	How to monitor the operation of the attachment and the actions that might be required
32	How to maintain the stability of the machine during unloading and driving & check the accuracy of the required assortment lengths and the correctness of assortment diameters
33	The slope limitations on the loading and unloading operations
34	The methods of loading and unloading long logs, poles or tree lengths & knows the difference between wood for wood and wood for pulp and wood
35	The lifting and reach capabilities and height limitations of the machinery used when unloading
36	The capabilities and limitations of the machinery used including its slope limitations & how to cross transverse ditches and riverbeds
37	How to maintain the stability of the machine during unloading and driving
38	How to segregate and grade products to meet the required specification when unloading
39	The safe stacking heights, stability of stacks and signing requirements

40	The use of bearers where appropriate when unloading
41	The potential impact of your work on the environment and how this can be minimised including the consequences of damage to standing trees with forestry machinery
Competence Level 5-Forwarder post-use inspection	
42	The reasons why the base machine and attachment should be left in a suitable condition for future use
43	Your responsibilities under relevant environmental, health and safety legislation and codes of practice &, if necessary, explain the purpose of their work with expert arguments and understand professional terminology

APPENDIX B: Data Set 3

Table. 23 – Mapping document 9-UK, CZ and SI Forestry Tractor/Skidder standards evaluation (Robb 2019)

UK Outcome	European Occupational Standards- Forestry Tractor/Skidder (UK)	CZ Outcome	SI Outcome
PERFORMANCE CRITERIA-you must be able to:			
1,4(m)	Assess the risks associated with the site and the proposed work & plan the work	1	1
2	Select and implement appropriate working method in accordance with the assessed risks	1,2	1
3	Select and use appropriate personal protective equipment (PPE) for the work		2
4,5/8(m)	Select & maintain the security of machinery and equipment on site and if necessary, safely load/unload from transporter		2
5,7(m)	Select, prepare and use the appropriate tools and equipment, safely and effectively		2
6(b/h)	Confirm all equipment has been checked and is fit for purpose (including attachments)		2
7(b/h)	Carry out routine operator maintenance, pre-start checks (includes safety guards) and the setting of the machine for use (includes operator environment) including replacement of hydraulic hose as necessary & fuel/lubricant checks	4	2,3,7

8(b)	Check the immediate work area for hazards and obstacles before moving off (use appropriate warning signals at all times)		3
9	Select, prepare and maintain access and egress routes in a serviceable condition		3
10(b)	Operate and manoeuvre the base machine safely, and in a way that is consistent with the type of vehicle, attachment and operation		3
11(b)	Make sure the attachment is in transport position when not in use, in accordance with specifications		3
12(b)	Modify operating procedures to take into account any changes in weather, ground conditions, types of terrain and site restraints		1,3
13(b)	Utilise additional safeguards when reversing or manoeuvring and comply with safety distances adjacent to roads and tracks or where others are working		3
14(b)	Maintain the efficiency of base machine and attachment performance through the appropriate handling and maintenance	1	3
15(b)	Deal with any hazards and obstacles encountered during the operation, in accordance with standard practice		3
16(b)	Monitor the operation of the attachment and recognise any faults		3

17	Ensure that damage to the worksite, any remaining standing trees, tracks, roads, drains and the wider environment is kept within specified limits		3
18(m),12(w), 8(s)	Identify, select & extract wood and wood products effectively and efficiently single and multiple felled trees, poles or other timber using agreed extraction routes	3	3
9(s)	position the machine and attach/load timber safely	1	f
9(w)	Ensure the cable and other system components are of adequate capacity and length for the load to be applied and the distance to be pulled		f
10(w)	maintain effective teamwork and communication with other team members when winching, using an appropriate method		f
11(w)	Pull out the cable, avoiding obstacles to the winching process		f
13(w)	Ensure timber is choked appropriately for safe and effective winching		f
14(w)	Operate mechanically powered winch to extract timber in keeping with site constraints, in a safe and effective way	1	f
24,16/17(m)	Grade, separate and accumulate products to facilitate their subsequent handling, processing or uplift for onward dispatch	3	3
15(m)	Stop winching when signalled to do so		
16(m)	Utilise additional safeguards and comply with safety distances when		f

	winching near roads and tracks or where others are working		
17(m)	Ensure that the timber is in a safe position before releasing chokers, or attaching the load to the winch, if skidding		
10(s)	Uplift timber to a capacity that ensures safe and effective skidding		f
11(s)	Use the skidder attachment to bunch products safely and effectively		f
12(s)	Uplift timber flush to the machine, as appropriate for the type of attachment		
13(s)	Limit the size of load on the skidder to avoid damage to the product, instability of the machine, loss of load, or damage to standing trees, tracks, roads, drains and the wider environment	1	
14(s)	Position the timber in the correct place for subsequent handling and detach/unload the timber from the machine safely and effectively at the accumulation point		f
25	Meet the relevant legislative and environmental requirements relevant to operating the machine, as well as those of your organisation		f
26(b)	Stop and isolate the base machine on completion of the activity, in line with standard practice		
27(b)	Remove the attachment, where appropriate, in accordance with specifications		

28(b)	Leave the base machine and attachment secure after use and in a suitable condition for future use		
29	Ensure the site is left in a condition which meets environmental requirements, in accordance with the specification		9
30	Maintain the health and safety of yourself and others at all times, in accordance with relevant legislation	1	9
KNOWLEDGE & UNDERSTANDING-you need to know & understand:			
1,6(m)	How to identify hazards and assess risks and importance of planning the work	1	9
2	How to interpret risk assessments		5
3	The selection, use and care of personal protective equipment (PPE)		9
4	Site procedures including communication systems and emergency planning		8
5	The types of tools and equipment required and how to maintain and use these safely and effectively in line with the manufacturer's recommendations	1,4	7
6	The legal requirements for checking equipment		3
7	Why it is important to maintain the safety and security of equipment and vehicles on site		2
8	The current guidelines on machinery operation, risk zones and safety clearances from overhead electricity conductors, working at height and		1,3

	what to do in the event of contact with power lines		
9(h/b)	The function of all operating controls for base unit, machine boom/ attachment etc	2,3,4	2,3
10,16(m)	The capabilities and limitations of the machine, in relation to slope limitations & maximum safe working load		3
11(b)	The required pre-start checks and adjustments for the base machine and attachments, and the reasons why these must be carried out	1	2
12(b)	The requirements for machine and operator protection structures		
13(b)	The procedure for loading and unloading base machine and attachments from transporter where appropriate	1	
14(b)	How to mount and dismount the base machine safely		
15(b)	How to start and stop the base machine including the correct procedures for jump starting		
16(b)	The types of attachments that are safe for use with the base machine and how they are fitted		
17(b)	The conditions that should be taken into account when considering the use of attachments		
18(b)	How to set or calibrate attachments, where required	4	
19(b)	The appropriate use of warning signals and indicators		

20(b),14(w)	The ways in which the base machine should be operated and manoeuvred, and how different weather, ground conditions, types of terrain and the work being carried out must be taken into account including when winching timber		1,3
21(b)	The various driving characteristics and techniques for driving a base machine with different wheel configurations		
22(b)	Correct fitting of traction aids and advantages and disadvantages of using them	2	
23(b)	How to operate and use attachments safely including the counterbalance		
24(b)	The controls and electrical services on the base machine and the attachment		
25(b)	How to put the attachment into transport position and work position, where appropriate	1	
26(b)	The capabilities of the base machine and the factors that may affect its efficiency		
27(b)	The types of hazards and obstacles that may be encountered and how these should be dealt with	1	f
28(b)	How to monitor the operation of the attachment and the actions that might be required		6
29(b)	The faults, defects and wearing of parts that occur with attachments used for treework operations		

30(b)	How to preserve the efficiency of the base machine and attachment through routine operator checks and basic maintenance in line with specifications	4	7
31(b)	How to carry out basic repairs and where to refer problems with the base machine or attachment	4	7
32(b)	The reasons why the base machine and attachment should be left in a suitable condition for future use		
33	The slope limitations on the loading and unloading operations		1,3,9
7(m)	How to select the right motorised machine and the advantages and disadvantages of using different machines to extract wood and wood products		
9(w)	Ensure that the timber is in a safe position before releasing chokers, or attaching the load to the winch, if skidding		f
10(w)	The acceptable limitations to cable wear when using the winch for assisted directional felling or the restraint of windblown trees		
11(w)	The information from the Certificate of Conformity required to ensure that the cable is of a stated breaking strain		
12(w)	The operator checks and maintenance necessary on the cable and its terminal, strops, pulleys, shackles, hooks, chokers etc. before and during		

	use, including identification of the load bearing capacity of the equipment		
14(w)	The additional problems, hazards and risks posed by working on steep slopes when winching timber		
15(w)	How to recognise the type of timber and the species and select the product categories in order to meet the required specification		
16(w)	What to look out for when routing the cable, with regard to obstacles that may be encountered when the winch is operated		
17(w)	The additional points to consider when laying out the cable and chokering multiple stems		
18(w)	What to consider when selecting strops and anchor points when setting pulleys used in offset pulling, double rigging or hi-leading, with regard to the multiplication of forces applied		
19(w)	The bearing capacity of pulley anchor strops in various configurations		
20(w)	The signals and methods of signalling between the winch operator or intermediate banksman and the chokerman		
21(w)	How to deal with timber that encounters an obstacle while it is being winched in		
22(w)	The effect of poorly choked or awkwardly shaped loads on line wire extraction		

23(w)	The additional hazards and risks involved when using vehicle-mounted winches with no ground anchors, such as on forestry tractors or skidders, and their unsuitability for restraint of windblown stems and root-plates or in assisted directional felling		f
24(w)	How offset pulleys should be set if winching down a steep slope, to avoid damage to winching equipment or injury to the operator		
36	How to maintain the stability of the machine during unloading and driving		9
37,15(m)	How to segregate and grade products to meet the required specification when unloading		3.7 (less specific)
38,17(s)	The safe stacking heights, stability of stacks and signing requirements		3
17/19(m)	How to select suitable extraction routes & effect of unbalanced loads on extraction	1	
14(s)	The implications of the load moving or falling when detaching/unloading		
15(s)	How and when to use bearers when unloading/detaching load		
16(s)	How and when to use a logging blade or butt plate to accumulate extracted timber		
40(b)	How to set up and use fuelling and maintenance areas where appropriate		
41	The potential impact of your work on the environment and how this can be minimised		9

42	Your responsibilities under relevant environmental, health and safety legislation and codes of practice		9
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Table. 24 – Mapping document 10-Comparative analysis of UK, CZ and SI Forestry Tractor/Skidder standards-additional findings (Robb 2019)

CZ Outcome	European Occupational Standards- Forestry Tractor/Skidder (CZ)	UK Outcome	SI Outcome
ADDITIONAL PERFORMANCE CRITERIA-you must be able to:			
d(l)	Extract logs and assortments on the transport site using universal or special forestry tractors, or other mechanization means		
d(l)	Work quality-prevent damage to soil, trees and their root systems during extraction		
SI Outcome	European Occupational Standards- Forestry Tractor/Skidder (SI)	UK Outcome	CZ Outcome
ADDITIONAL PERFORMANCE CRITERIA-you must be able to:			
f	Get acquainted with the situation on the ground		
1, f	Get acquainted with the tasks in the harvesting units and on the job site with the help of a work order, harvesting plan or contract or oral instructions regarding the classification of forest-timber assortments		
2, f	Check the suitability and amount of fuel and lubricants in the machine		
8, f	Communicates with co-workers with gestures or verbal communication to ensure safety at work		

f	Perform an empty ride with the tractor in the correct position		
3, f	Removes larger branches, bark and debris/residues the road during unloading so as not to obstruct traffic		
f	Repairs damage to the forest road		
f	Selects the anchorage location of the tractor and the direction of attracting wood to the road		
f	Uses a mechanical and remote-controlled winch		
f	Ergonomically stretches the tow rope and binds assortments		
f	Visually monitors assortments along the path of attraction and eliminates congestion and delays in attraction		
f	Ergonomically stretches the winch cable and binds assortments		
f	stacks conifer branches and tops in piles and cuts thicker deciduous branches		
f	Clean and keep waterways, ditches etc clear		
f	Present and implement measures for safe work with a tractor winch in determining the strength of dry and rotten wood		
f	Assists the logger forester in felling heavily leaning and hung-up trees		
f	Demonstrates harvesting and felling in hazardous conditions		
KNOWLEDGE & UNDERSTANDING-you need to know & understand:			
f	Anticipate dangerous situations according to the direction and method		

	of harvesting: snow, fire, ice, or wind damage conditions		
f	Knows the advantages and disadvantages of different types of soil that affect harvesting with a tractor		
3, f	Knows the difference between wood for wood and wood for pulp and wood & classifies assortments into logs, wood for boards and pulp and firewood		
3, f	Knows the consequences of damage to standing trees with forestry machinery		
f	Knows the dangers that can arise from incorrectly positioned tractors		
3, f	Knows the dangers of driving with a full load or at speed		

Table. 25 – Mapping document 11a-Thematic analysis of UK Forestry Tractor/Skidder standards (Robb 2019)

Theme	Forestry Tractor/Skidder & Base Machine
Country	United Kingdom
Standard Name and Code	Operate a grapple loader (code: LANTw47), Prepare and operate a base machine with attachments (code: LANTw79), Extract wood and wood products using motorised machines (code: LANTw43), Extract timber using mechanically powered winches (code: LANTw42) & Attach and detach timber onto a skidder (code: LANTw46)
Responsible authority	Skills Development Scotland
Validity of the standard	From February 2017
Qualification level	EQF (undefined) although promoted by providers at level 3

Name/Scope of professional competence level	Forestry tractor /skidder 1-Use one or more of the following: • tractors • ATV adaptations • compact skidders and "iron horse" • similar tracked machines 2-Timber, either as single or multiple stems, both tip first and butt first 3-Communicate with other team members using one or more of the following: • hand signals • two-way radio • headsets
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Table. 26 – Mapping document 11b-Thematic analysis of CZ Forestry Tractor/Skidder standards (Robb 2019)

Theme	Harvester & Forwarder and Forestry tractor
Country	Czech Republic
Standard Name and Code	Timber harvesting with logging and transport machines (code: 41-091-M) & Logging operations part d (code: NSK 41-023-H)
Responsible authority	Ministry of Agriculture
Validity of the standard	From 19.6.2020
Qualification level	EQF 3
Name/Scope of professional competence level	Forestry tractor/skidder 1-Extraction of logs and assortments on the transport site using universal or special tractors, or other mechanization means

Table. 27 – Mapping document 11c-Thematic analysis of SI Forestry Tractor/Skidder standards (Robb 2019)

Theme	Harvester & Forwarder
Country	Slovenia
Standard Name and Code	Operator / operator of machines for machine felling and export of wood with articulated semi-trailer (code: 78281880) & Forestry tractor (code: 6718282011)
Responsible authority	Ministry of Labour, Family, Social Affairs and Equal Opportunities
Validity of the standard	From 22.05.2017
Qualification level	EQF 4
Name/Scope of professional competence level	Forestry tractor/skidder 1-Inspects the harvesting area and performs an empty tractor drive 2- Carries out the collection of forest-wood assortments and full loaded drive of the tractor 3- Arranges the work site, loading area and implements the forest order 4- Participates in felling in special conditions using forestry machinery 5- Performs harvesting and felling of assortments in emergency conditions with various forestry machinery

Table. 28 – Mapping document 12-Harmonised European Forestry Tractor/Skidder standards (Robb 2019)

Outcome	European Occupational Standards-Forestry Tractor/Skidder
PERFORMANCE CRITERIA-you must be able to:	
Competence Level 1-Work Planning & Preparation	

1	Assess the risks associated with the site and the proposed work & identify basic forest tree species/read forestry maps/plans as applicable
2	Select and implement appropriate working method in accordance with the assessed risks
3	Select and use appropriate personal protective equipment (PPE) for the work
4	Maintain the security of machinery and equipment on site
5	Select & maintain the security of machinery and equipment on site and if necessary, safely load/unload from transporter
6	Select, prepare and use the appropriate tools and equipment, safely and effectively
7	Get acquainted with the situation on the ground, the tasks in the harvesting units and on the job site with the help of a work order, harvesting plan or contract or oral instructions regarding the classification of forest-timber assortments
Competence Level 2-Base machine pre-use inspection & maintenance	
8	Make sure the attachment is in transport position when not in use, in accordance with specifications
9	Carry out routine operator maintenance, pre-start checks (includes safety guards) and the setting of the machine for use (includes operator environment) including replacement of hydraulic hose as necessary & fuel/lubricant checks
Competence Level 3- Forestry Tractor/Skidder safety checks	
10	Confirm all equipment has been checked and is fit for purpose including attachments
11	Check the immediate work area for hazards and obstacles before moving off (use appropriate warning signals at all times)
Competence Level 4- Forestry Tractor/Skidder operations	
12	Select, prepare and maintain access and egress routes in a serviceable condition
13	Operate and manoeuvre the base machine safely, and in a way that is consistent with the type of vehicle, attachment and operation

14	Modify operating procedures to take into account any changes in weather, ground conditions, types of terrain and site restraints
15	Utilise additional safeguards when reversing or manoeuvring and comply with safety distances adjacent to roads and tracks or where others are working
14	Maintain the efficiency of base machine and attachment performance through the appropriate handling and maintenance
15	Deal with any hazards and obstacles encountered during the operation, in accordance with standard practice & communicate with co-workers with gestures or verbal communication to ensure safety at work
16	Monitor the operation of the attachment and recognise any faults
17	Ensure that damage to the worksite, any remaining standing trees, tracks, roads, drains and the wider environment is kept within specified limits
18	Identify, select & extract wood and wood products effectively and efficiently single and multiple felled trees, poles or other timber using agreed extraction routes
19	Position the machine and attach/load timber safely
20	Ensure the cable and other system components are of adequate capacity and length for the load to be applied and the distance to be pulled
21	Maintain effective teamwork and communication with other team members when winching, using an appropriate method
22	Pull out the cable, avoiding obstacles to the winching process
23	Ensure timber is choked appropriately for safe and effective winching
24	Operate mechanically powered winch to extract timber in keeping with site constraints, in a safe and effective way
25	Grade, separate and accumulate products to facilitate their subsequent handling, processing or uplift for onward dispatch
26	Stop winching when signalled to do so

27	Utilise additional safeguards and comply with safety distances when winching near roads and tracks or where others are working
28	Ensure that the timber is in a safe position before releasing chokers, or attaching the load to the winch, if skidding
29	Uplift timber to a capacity that ensures safe and effective skidding
30	Use the skidder attachment to bunch products safely and effectively
31	Uplift timber flush to the machine, as appropriate for the type of attachment
32	Limit the size of load on the skidder to avoid damage to the product, instability of the machine, loss of load, or damage to standing trees, tracks, roads, drains and the wider environment
33	Position the timber in the correct place for subsequent handling and detach/unload the timber from the machine safely and effectively at the accumulation point
34	Communicates with co-workers with gestures or verbal communication to ensure safety at work
35	Perform an empty ride with the tractor in the correct position
36	Removes larger branches, bark and debris/residues the road during unloading so as not to obstruct traffic
37	Repairs damage to the forest road
38	Selects the anchorage location of the tractor and the direction of attracting wood to the road
39	Uses a mechanical and remote-controlled winch
40	Ergonomically stretches the tow rope and binds assortments
41	Visually monitors assortments along the path of attraction and eliminates congestion and delays in attraction
42	Ergonomically stretches the winch cable and binds assortments
43	Stacks conifer branches and tops in piles and cuts thicker deciduous branches
44	Clean and keep waterways, ditches etc clear

45	Present and implement measures for safe work with a tractor winch in determining the strength of dry and rotten wood
46	Assists the logger forester in felling heavily leaning and hung-up trees
47	Demonstrates harvesting and felling in hazardous conditions
48	Grade, separate and accumulate products to facilitate their subsequent handling, processing or uplift for onward dispatch
49	Meet the relevant legislative and environmental requirements relevant to operating the machine, as well as those of your organisation
Competence Level 5- Forestry Tractor/Skidder post-use inspection	
50	Stop and isolate the base machine on completion of the activity, in line with standard practice
51	Remove the attachment, where appropriate, in accordance with specifications
52	Leave the base machine and attachment secure after use and in a suitable condition for future use
53	Ensure the site is left in a condition which meets environmental requirements, in accordance with the specification
54	Maintain the health and safety of yourself and others at all times, in accordance with relevant legislation
KNOWLEDGE & UNDERSTANDING-you need to know & understand:	
Competence Level 1-Work Planning & Preparation	
1	How to identify hazards and assess risks and importance of planning the work
2	How to interpret risk assessments
3	The types of hazards and obstacles that may be encountered and how these should be dealt with
4	The selection, use and care of personal protective equipment (PPE)
5	Site procedures including communication systems and emergency planning & knows the rules of teamwork

6	Why it is important to maintain the safety and security of equipment and vehicles on site & understands costs of fuel, labour and materials etc
7	The current guidelines on machinery operation, risk zones and safety clearances from overhead electricity conductors, working at height and what to do in the event of contact with power lines
8	How to set up and use fuelling and maintenance areas where appropriate
9	The types of tools and equipment required and how to maintain and use these safely and effectively in line with the manufacturer's recommendations
Competence Level 2-Base machine pre-use inspection & maintenance	
10	The procedure for loading and unloading base machine and attachments from transporter where appropriate
11	The legal requirements for checking equipment
12	How to mount and dismount the base machine safely
13	How to start and stop the base machine including the correct procedures for jump starting
14	The types of attachments that are safe for use with the base machine and how they are fitted
15	The conditions that should be taken into account when considering the use of attachments
	How to set or calibrate attachments, where required
16	The requirements for machine and operator protection structures
17	Correct fitting of traction aids and advantages and disadvantages of using them
18	How to preserve the efficiency of the base machine and attachment through routine operator checks and basic maintenance in line with specifications
19	How to carry out basic repairs and where to refer problems with the base machine or attachment & keep records including machine hours etc

20	The controls and electrical services on the base machine and the attachment
21	The faults, defects and wearing of parts that occur with attachments used for treework operations
22	The function of all operating controls for base unit, machine boom/attachment etc
23	The operator checks and maintenance necessary on the cable and its terminal, strops, pulleys, shackles, hooks, chokers etc. before and during use, including identification of the load bearing capacity of the equipment
24	The acceptable limitations to cable wear when using the winch for assisted directional felling or the restraint of windblown trees
25	The information from the Certificate of Conformity required to ensure that the cable is of a stated breaking strain
26	What to consider when selecting strops and anchor points when setting pulleys used in offset pulling, double rigging or hi-leading, with regard to the multiplication of forces applied
27	The bearing capacity of pulley anchor strops in various configurations
Competence Level 3- Forestry Tractor/Skidder safety checks	
28	The required pre-start checks and adjustments for the base machine and attachments, and the reasons why these must be carried out
29	How to select the right motorised machine (universal/forestry tractors) and the advantages and disadvantages of using different machines to extract wood and wood products
Competence Level 4- Forestry Tractor/Skidder operations	
24	The appropriate use of warning signals and indicators
25	The ways in which the base machine should be operated and manoeuvred, and how different weather, ground conditions, types of terrain and the work being carried out must be taken into account
26	The various driving characteristics and techniques for driving a base machine with different wheel configurations

27	How to operate and use attachments safely including the counterbalance
28	How to put the attachment into transport position and work position, where appropriate
29	The capabilities of the base machine and the factors that may affect its efficiency
30	The capabilities and limitations of the machine when loading, in relation to maximum safe working load
31	How to monitor the operation of the attachment and the actions that might be required
32	How to maintain the stability of the machine during unloading and driving & check the accuracy of the required assortment lengths and the correctness of assortment diameters
33	The slope limitations on the loading and unloading operations
34	The signals and methods of signalling between the winch operator or intermediate banksman and the chokerman
35	Ensure that the timber is in a safe position before releasing chokers, or attaching the load to the winch, if skidding
36	The additional points to consider when laying out the cable and chokering multiple stems
37	The additional problems, hazards and risks posed by working on steep slopes when winching timber
38	How to recognise the type of timber and the species and select the product categories in order to meet the required specification
39	How to segregate and grade products to meet the required specification when unloading
40	What to look out for when routing the cable, with regard to obstacles that may be encountered when the winch is operated
41	How to deal with timber that encounters an obstacle while it is being winched in
42	The potential impact of your work on the environment and how this can be minimised including the consequences of damage to standing trees/root systems or soil with forestry extraction machinery

43	How to deal with timber that encounters an obstacle while it is being winched in
44	The effect of poorly choked or awkwardly shaped loads on line wire extraction
45	The additional hazards and risks involved when using vehicle-mounted winches with no ground anchors, such as on forestry tractors or skidders, and their unsuitability for restraint of windblown stems and root-plates or in assisted directional felling
46	How offset pulleys should be set if winching down a steep slope, to avoid damage to winching equipment or injury to the operator
47	The safe stacking heights, stability of stacks and signing requirements
48	How to select suitable extraction routes & effect of unbalanced loads on extraction
49	The implications of the load moving or falling when detaching/unloading
50	How and when to use bearers when unloading/detaching load
51	How and when to use a logging blade or butt plate to accumulate extracted timber
52	Anticipate dangerous situations according to the direction and method of harvesting: snow, fire, ice, or wind damage conditions
53	Knows the advantages and disadvantages of different types of soil that affect harvesting with a tractor
54	Knows the difference between wood for wood and wood for pulp and wood & classifies assortments into logs, wood for boards and pulp and firewood
55	Knows the consequences of damage to standing trees with forestry machinery
56	Knows the dangers that can arise from incorrectly positioned tractors
57	Knows the dangers of driving with a full load or at speed
Competence Level 5- Forestry Tractor/Skidder post-use inspection	

58	The reasons why the base machine and attachment should be left in a suitable condition for future use
59	Your responsibilities under relevant environmental, health and safety legislation and codes of practice &, if necessary, explain the purpose of their work with expert arguments and understand professional terminology

